

#1 · Probability Models

What is the pdf of the exponential distribution?

Answer

$f(x) = (1/\theta) \times \exp(-x/\theta)$ for $x > 0$, where θ is the scale parameter and also equals the mean.

#2 · Probability Models

What does the memoryless property of the exponential distribution mean?

Answer

Given a loss has already exceeded d , the distribution of the excess over d is exponential with the same parameter θ . Past survival gives no information about the future.

#3 · Probability Models

What is the hazard rate of the exponential distribution?

Answer

A constant, $h(x) = 1/\theta$ for all x . This constant hazard rate is what produces the memoryless property.

#4 · Probability Models

What is the mean excess function $e(d)$ for the exponential distribution?

Answer

$e(d) = \theta$ for every d , a direct consequence of the memoryless property.

#5 · Probability Models

What is a scale parameter?

Answer

A parameter that stretches or compresses a distribution along the x-axis without changing its shape; multiplying a random variable by a constant c multiplies its scale parameter by c .

#6 · Probability Models

What is a shape parameter?

Answer

A parameter that controls the qualitative form of the distribution, such as skewness and tail weight, rather than just stretching it. Changing it changes the distribution's behavior, not just its scale.

#7 · Probability Models

Why does scale invariance matter for trending losses?

Answer

Because trending or inflating losses by a constant factor simply multiplies the scale parameter by that factor, leaving the shape parameter unchanged, which makes adjusting severity distributions for inflation straightforward.

#8 · Probability Models

Give the survival function of the single-parameter Pareto distribution.

Answer

$S(x) = (\theta/x)^{\alpha}$ for $x \geq \theta$, where α is the shape parameter and θ is a fixed lower threshold.

#9 · Probability Models

Give the pdf of the two-parameter Pareto (Pareto Type II) distribution.

Answer

$f(x) = (\alpha \times \theta^\alpha) \div (x + \theta)^{(\alpha+1)}$ for $x > 0$, with θ as the scale parameter and α as the shape parameter.

#10 · Probability Models

What is the mean of the two-parameter Pareto distribution, and when does it exist?

Answer

$E[X] = \theta \div (\alpha - 1)$, which exists only when $\alpha > 1$. For $\alpha \leq 1$ the mean is infinite or undefined.

#11 · Probability Models

When does the variance of the two-parameter Pareto exist?

Answer

Only when $\alpha > 2$. As α decreases toward 1, the tail becomes heavier and even the mean can fail to exist.

#12 · Probability Models

What is the mean excess function of the two-parameter Pareto?

Answer

$e(d) = (\theta + d) \div (\alpha - 1)$, which is increasing in d , a hallmark of heavy-tailed behavior.

#13 · Probability Models

Why is an increasing mean excess function significant?

Answer

It signals a heavy-tailed distribution: the more a loss has already exceeded a threshold, the larger the expected further excess becomes, unlike light-tailed distributions where mean excess is flat or decreasing.

#14 · Probability Models

How does the tail of the two-parameter Pareto compare to the exponential?

Answer

The Pareto's tail decays polynomially (like a constant over x^{α}), which is heavier than the exponential's exponential decay, making large claims relatively more likely under the Pareto.

#15 · Probability Models

Why is the Pareto distribution favored for excess and reinsurance layers?

Answer

Because its heavy polynomial tail better reflects the real possibility of very large claims, which are exactly what drive the loss cost of a high excess layer.

#16 · Probability Models

Give the pdf of the lognormal distribution.

Answer

$f(x) = \frac{1}{x \times \sigma \times \sqrt{2 \times \pi}} \times \exp\left(-\frac{(\ln(x) - \mu)^2}{2 \times \sigma^2}\right)$ for $x > 0$, where $\ln(X)$ is normal with mean μ and standard deviation σ .

#17 · Probability Models

What is $E[X]$ for a lognormal distribution with parameters μ and σ ?

Answer

$$E[X] = \exp(\mu + \sigma^2/2).$$

#18 · Probability Models

Which parameter of the lognormal controls tail weight and skewness?

Answer

σ , the standard deviation of $\ln(X)$. Larger σ produces a more heavily skewed distribution with a heavier right tail.

#19 · Probability Models

How does the lognormal's tail compare to the Pareto's?

Answer

The lognormal's tail is lighter than the Pareto's; all moments of the lognormal exist, whereas Pareto moments can fail to exist for small α .

#20 · Probability Models

How do you compute $P(X \leq x)$ for a lognormal random variable?

Answer

$P(X \leq x) = \Phi((\ln(x) - \mu) / \sigma)$, using the standard normal cdf Φ , since $\ln(X)$ is normally distributed.

#21 · Probability Models

Give the pdf of the gamma distribution.

Answer

$f(x) = (x^{(\alpha-1)} \times \exp(-x \div \theta)) \div (\theta^\alpha \times \Gamma(\alpha))$ for $x > 0$, with shape parameter α and scale parameter θ .

#22 · Probability Models

What are the mean and variance of the gamma distribution?

Answer

Mean = $\alpha \times \theta$, Variance = $\alpha \times \theta^2$.

#23 · Probability Models

What happens to the gamma distribution when $\alpha = 1$?

Answer

It reduces exactly to the exponential distribution with parameter θ .

#24 · Probability Models

How does the gamma density behave near $x = 0$ when $\alpha > 1$, compared to the exponential?

Answer

Unlike the exponential (always decreasing from $x = 0$), the gamma density starts at 0 when $\alpha > 1$ and rises to a peak before decreasing.

#25 · Probability Models

Is the gamma distribution heavy-tailed or light-tailed?

Answer

Light-to-moderate tailed. Its tail decays exponentially, unlike the polynomial tail decay of the Pareto.

#26 · Probability Models

Give the cdf of the Weibull distribution.

Answer

$F(x) = 1 - \exp(-(x/\theta)^\tau)$ for $x > 0$, with shape parameter τ and scale parameter θ .

#27 · Probability Models

What happens to the Weibull distribution when $\tau = 1$?

Answer

It reduces to the exponential distribution with parameter θ .

#28 · Probability Models

How does the Weibull's tail and hazard rate behave when $\tau < 1$?

Answer

The tail is heavier than exponential and the hazard rate is decreasing in x .

#29 · Probability Models

How does the Weibull's tail and hazard rate behave when $\tau > 1$?

Answer

The tail is lighter than exponential and the hazard rate is increasing in x .

#30 · Probability Models

Why are the gamma and Weibull distributions useful for hypothesis testing against the exponential?

Answer

Both nest the exponential as a special case ($\alpha = 1$ or $\tau = 1$), so a likelihood ratio test can check whether the added shape flexibility is actually needed.

#31 · Probability Models

What is a finite mixture distribution?

Answer

A distribution formed by combining two or more component densities with mixing weights that sum to 1, e.g. $f(x) = p \times f_1(x) + (1-p) \times f_2(x)$.

#32 · Probability Models

How is the mean of a two-component mixture computed?

Answer

$E[X] = p \times E[X_1] + (1-p) \times E[X_2]$, a simple weighted average of the component means.

#33 · Probability Models

Is the variance of a mixture equal to the weighted average of the component variances?

Answer

No. It equals the weighted average of the component second moments minus the square of the mixture mean, which adds an extra between-component variance term whenever the component means differ.

#34 · Probability Models

What law explains why mixture variance exceeds the weighted average of component variances?

Answer

The law of total variance: $\text{Var}(X) = E[\text{Var}(X|\text{component})] + \text{Var}(E[X|\text{component}])$. The second term adds dispersion whenever component means differ.

#35 · Probability Models

Why might an actuary use a mixture of an exponential and a Pareto?

Answer

To capture two distinct claim types in one model: the exponential for many small, routine claims and the Pareto for a smaller share of severe, heavy-tailed claims.

#36 · Probability Models

What is the coefficient of variation (CV)?

Answer

$CV = \text{standard deviation} \div \text{mean}$. It measures relative dispersion and allows comparison of variability across distributions with different means.

#37 · Probability Models

What is the CV of the exponential distribution?

Answer

Exactly 1, regardless of θ , since the standard deviation always equals the mean.

#38 · Probability Models

What does a CV greater than 1 generally indicate?

Answer

A heavier tail than the exponential distribution; CV less than 1 generally indicates a lighter, more concentrated tail.

#39 · Probability Models

Define the limited expected value function $E[X \wedge d]$.

Answer

The expected value of a loss capped at d : $E[X \wedge d]$
= integral from 0 to d of $S(x)$ dx . It equals the expected payment under a policy limit of d .

#40 · Probability Models

What happens to $E[X \wedge d]$ as d approaches ∞ ?

Answer

It approaches $E[X]$, the unconditional mean, assuming the mean exists.

#41 · Probability Models

Give the closed-form limited expected value for the two-parameter Pareto.

Answer

$$E[X \wedge d] = (\theta \div (\alpha - 1)) \times [1 - (\theta \div (\theta + d))^{\alpha - 1}].$$

#42 · Probability Models

Define the mean excess (mean residual life) function $e(d)$.

Answer

$e(d) = E[X - d \mid X > d]$, the expected amount by which a loss exceeds d , given it has already exceeded d . It represents the average claim size in an excess layer attaching at d .

#43 · Probability Models

Define the hazard rate function $h(x)$.

Answer

$h(x) = f(x) \div S(x)$, the instantaneous rate at which losses of size x terminate given they have survived to x .

#44 · Probability Models

What does a decreasing hazard rate imply about tail weight?

Answer

A decreasing hazard rate corresponds to a heavier tail than the exponential, since large losses become relatively more likely, not less, as x grows.

#45 · Probability Models

What does an increasing hazard rate imply about tail weight?

Answer

An increasing hazard rate corresponds to a lighter tail than exponential, since large losses become relatively less likely as x grows.

#46 · Probability Models

Write the general log-likelihood for complete (uncensored, untruncated) loss data.

Answer

$l(\theta) = \sum_i \ln(f(x_i; \theta))$, the sum of the log-density evaluated at each observed loss.

#47 · Probability Models

What is censored data in a P&C loss context?

Answer

Data where a loss is known only to be at least a certain value, most commonly because a policy limit caps the recorded loss at u even though the true loss may have been larger.

#48 · Probability Models

What is the likelihood contribution of a right-censored observation at u ?

Answer

$S(u; \theta)$, the survival probability at u , not the density $f(u; \theta)$, since all that is known is the true loss exceeded u .

#49 · Probability Models

What is truncated data in a P&C loss context?

Answer

Data where losses below a threshold (e.g. a deductible d) are never recorded at all, so the observed sample only contains losses conditional on exceeding d .

#50 · Probability Models

What is the likelihood contribution of an observation left-truncated at d ?

Answer

$f(x_i; \theta) \div S(d; \theta)$, the density divided by the probability of exceeding the truncation point, since the observation is drawn from the conditional distribution given $X > d$.

#51 · Probability Models

What happens if you fit an unconditional density to truncated data without adjusting for truncation?

Answer

The fit is systematically biased: it typically underestimates the mean and overstates how small claims tend to be, because the entire lower tail below the deductible is missing from the sample.

#52 · Probability Models

Write the log-likelihood term for an observation that is both left-truncated at d and censored at limit u .

Answer

$\ln(S(u; \theta) \div S(d; \theta))$, combining the truncation adjustment (dividing by $S(d)$) with the censoring adjustment (using the survival probability at u instead of the density).

#53 · Probability Models

For an exponential distribution fit by MLE to complete data, what is θ_{hat} ?

Answer

θ_{hat} equals the sample mean of the observed losses, since setting the derivative of the log-likelihood to zero gives $\theta_{\text{hat}} = \text{sum}(x_i) \div n$.

#54 · Probability Models

Five claims of 200, 350, 500, 150, 800 are observed in full. What is the MLE of θ for an exponential fit?

Answer

$\theta_{\text{hat}} = \text{sample mean} = 2,000 \div 5 = 400$.

#55 · Probability Models

What is the method of moments estimation approach?

Answer

Set the theoretical moments of a distribution (mean, variance, etc.) equal to the corresponding sample moments and solve the resulting equations for the parameters.

#56 · Probability Models

For a gamma distribution, what two equations does method of moments solve?

Answer

$\alpha \times \theta = \text{sample mean}$ and $\alpha \times \theta^2 = \text{sample variance}$, solved simultaneously for α_{hat} and θ_{hat} .

#57 · Probability Models

What is a key disadvantage of method of moments compared to MLE?

Answer

It generally produces less efficient estimators (higher variance) and does not naturally handle censored or truncated data the way the likelihood framework does.

#58 · Probability Models

Why is MLE preferred over method of moments in P&C severity modeling?

Answer

MLE is generally more efficient and its likelihood framework naturally adjusts for censored and truncated data, both very common in real loss data, whereas method of moments does not.

#59 · Probability Models

Losses follow a two-parameter Pareto with $\alpha = 3$, $\theta = 1,000$. What is $E[X \wedge 2000]$?

Answer

$$E[X \wedge 2000] = 500 \times [1 - (1000 \div 3000)^2] = 500 \times (8 \div 9) = 444.44.$$

#60 · Probability Models

In the Pareto example with $\alpha = 3$, $\theta = 1,000$, how much expected loss does a \$2,000 policy limit remove compared to the unlimited mean?

Answer

The unlimited mean is $\theta \div (\alpha - 1) = 500$. With the limit, $E[X \wedge 2000] = 444.44$, so the limit removes 55.56 of expected loss per claim.

#61 · Probability Models

Why is a heavy tail dangerous to underestimate in ratemaking?

Answer

Using too light a distribution underprices excess layers and leaves reserves inadequate for large claims, since the model understates how likely extreme losses really are.

#62 · Probability Models

Which distributions in this chapter nest the exponential as a special case?

Answer

The gamma (at $\alpha = 1$) and the Weibull (at $\tau = 1$) both reduce exactly to the exponential distribution.

#63 · Probability Models

Rank the exponential, gamma, lognormal, and two-parameter Pareto roughly from lightest to heaviest tail.

Answer

Exponential and gamma are light-to-moderate tailed (exponential decay), the lognormal is moderately heavy-tailed (all moments exist), and the two-parameter Pareto is the heaviest tailed (polynomial decay, moments can fail to exist).

#64 · Probability Models

What is the pmf of a Poisson distribution with parameter lambda?

Answer

$P(N=k) = \exp(-\lambda) \times \lambda^k \div k!$, for $k = 0, 1, 2, \dots$

#65 · Probability Models

What are the mean and variance of a Poisson distribution with parameter λ ?

Answer

Both equal λ : $E[N] = \lambda$ and $\text{Var}(N) = \lambda$. This is called equidispersion.

#66 · Probability Models

If $N_1 \sim \text{Poisson}(\lambda_1)$ and $N_2 \sim \text{Poisson}(\lambda_2)$ are independent, what is the distribution of $N_1 + N_2$?

Answer

$N_1 + N_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$. The Poisson family is closed under independent summation.

#67 · Probability Models

What is the probability generating function of a Poisson(lambda) random variable?

Answer

$$P(z) = E[z^N] = \exp(\text{lambda}(z-1))$$

#68 · Probability Models

What is the pmf of a binomial distribution with parameters m and q?

Answer

$$P(N=k) = C(m,k) \times q^k \times (1-q)^{(m-k)}, \text{ for } k = 0, 1, \dots, m$$

#69 · Probability Models

What are the mean and variance of a binomial distribution with parameters m and q ?

Answer

$E[N] = mq$ and $\text{Var}(N) = mq(1-q)$. Since $1-q < 1$, $\text{Var}(N)$ is always less than $E[N]$ (underdispersion).

#70 · Probability Models

When is the binomial distribution the natural choice for a frequency model?

Answer

When there is a hard, fixed upper bound m on the number of possible claims, such as the number of lives insured in a group policy.

#71 · Probability Models

What is the pmf of a negative binomial distribution with parameters r and β ?

Answer

$$P(N=k) = C(k+r-1, k) \times (1 \div (1+\beta))^r \times (\beta \div (1+\beta))^k, \text{ for } k = 0, 1, 2, \dots$$

#72 · Probability Models

What are the mean and variance of a negative binomial distribution with parameters r and β ?

Answer

$E[N] = r \times \beta$ and $\text{Var}(N) = r \times \beta \times (1+\beta)$. Since $1+\beta > 1$, $\text{Var}(N)$ always exceeds $E[N]$ (overdispersion).

#73 · Probability Models

Why is the negative binomial distribution often preferred over the Poisson distribution for real claim count data?

Answer

Real claim data usually shows variance greater than the mean (overdispersion), which the Poisson distribution's equidispersion property cannot capture, but the negative binomial can.

#74 · Probability Models

How can the negative binomial distribution be derived as a mixture?

Answer

As a Poisson-gamma mixture: if λ is gamma-distributed with shape r and scale β , and $N|\lambda$ is $\text{Poisson}(\lambda)$, then the unconditional distribution of N is negative binomial(r, β).

#75 · Probability Models

What does overdispersion mean for a claim count variable?

Answer

Overdispersion means the observed variance of the claim count exceeds its observed mean, violating the Poisson equidispersion assumption.

#76 · Probability Models

Name two common causes of overdispersion in real claims data.

Answer

Unobserved heterogeneity across policyholders (rating variables do not capture every risk difference) and contagion (one claim event makes related claim events more likely).

#77 · Probability Models

What is the (a,b,0) class recursion?

Answer

$p_k \div p_{(k-1)} = a + b \div k$, for $k = 1, 2, 3, \dots$, where
 $p_k = P(N=k)$.

#78 · Probability Models

Which three distributions make up the (a,b,0) class?

Answer

Poisson, binomial, and negative binomial.

#79 · Probability Models

What are a and b for the $\text{Poisson}(\lambda)$ distribution in the $(a,b,0)$ class?

Answer

$a = 0$, $b = \lambda$, with $p_0 = \exp(-\lambda)$.

#80 · Probability Models

What are a and b for the $\text{binomial}(m,q)$ distribution in the $(a,b,0)$ class?

Answer

$a = -q/(1-q)$, $b = (m+1)q/(1-q)$, with $p_0 = (1-q)^m$.

#81 · Probability Models

What are a and b for the negative binomial(r, β) distribution in the $(a, b, 0)$ class?

Answer

$a = \beta \div (1 + \beta)$, $b = (r - 1) \times \beta \div (1 + \beta)$, with $p_0 = (1 + \beta)^{-r}$.

#82 · Probability Models

How can you quickly tell which $(a, b, 0)$ distribution you have just from the sign of a?

Answer

$a = 0$ means Poisson. a negative means binomial. a positive means negative binomial.

#83 · Probability Models

What is the $(a,b,1)$ class and how does it differ from $(a,b,0)$?

Answer

The $(a,b,1)$ class uses the same recursion $p_k/p_{k-1} = a + b/k$ starting at $k=1$ or later, but allows p_0 to be set independently of the base formula, used for zero-modified and zero-truncated distributions.

#84 · Probability Models

In the collective risk model, how is the aggregate loss S defined?

Answer

$S = X_1 + X_2 + \dots + X_N$, where N is a frequency random variable and $X_1, X_2, \dots, X_N$ are iid severity random variables independent of N .

#85 · Probability Models

What are the two key structural assumptions of the collective risk model?

Answer

The severities X_i are independent and identically distributed, and the X_i are independent of the frequency N .

#86 · Probability Models

What is a compound distribution?

Answer

The distribution of a random sum $S = X_1 + \dots + X_N$ under the collective risk model, written $S \sim \text{Compound}(N, X)$.

#87 · Probability Models

In the individual risk model, how is the aggregate loss S defined?

Answer

$S = Y_1 + Y_2 + \dots + Y_n$, where n is a fixed, known number of policies and each Y_i is the loss from policy i .

#88 · Probability Models

How does the individual risk model differ from the collective risk model in terms of the loss terms Y_i vs X_i ?

Answer

In the individual risk model the Y_i need not be identically distributed (different policies can differ), while in the collective risk model the X_i must be iid; also n is fixed in the individual model but N is random in the collective model.

#89 · Probability Models

When is the individual risk model more natural than the collective risk model?

Answer

When modeling a known, finite set of distinguishable policies with different coverage limits and expected frequencies, such as a commercial property portfolio with 500 named insureds.

#90 · Probability Models

What is the compound-variance formula for $E[S]$?

Answer

$$E[S] = E[N] \times E[X]$$

#91 · Probability Models

What is the compound-variance formula for $\text{Var}(S)$?

Answer

$$\text{Var}(S) = E[N] \times \text{Var}(X) + \text{Var}(N) \times (E[X])^2$$

#92 · Probability Models

In the compound-variance formula, what does the term $E[N] \times \text{Var}(X)$ represent?

Answer

The contribution to total variance from variability in individual claim sizes, scaled by the average number of claims.

#93 · Probability Models

In the compound-variance formula, what does the term $\text{Var}(N) \times (E[X])^2$ represent?

Answer

The contribution to total variance from variability in the number of claims, magnified by the square of the average claim size.

#94 · Probability Models

If $N \sim \text{Poisson}(\lambda=4)$ and X has mean 1,000 and variance 500,000, independent of N , what is $E[S]$?

Answer

$$E[S] = E[N] \times E[X] = 4 \times 1,000 = 4,000$$

#95 · Probability Models

If $N \sim \text{Poisson}(\lambda=4)$ and X has mean 1,000 and variance 500,000, independent of N , what is $\text{Var}(S)$?

Answer

$$\begin{aligned}\text{Var}(S) &= E[N] \times \text{Var}(X) + \text{Var}(N) \times (E[X])^2 = 4 \times 500,000 \\ &+ 4 \times 1,000^2 = 2,000,000 + 4,000,000 = 6,000,000\end{aligned}$$

#96 · Probability Models

Why can frequency variability dominate total aggregate loss variance even when $\text{Var}(N)$ looks small on its own?

Answer

Because the $\text{Var}(N)$ term in the compound-variance formula is multiplied by $(E[X])^2$, the square of the average claim size, which can be large even if $\text{Var}(N)$ itself is modest.

#97 · Probability Models

What condition must hold on the frequency distribution for Panjer's recursion to apply exactly?

Answer

The frequency distribution must belong to the $(a,b,0)$ class (Poisson, binomial, or negative binomial).

#98 · Probability Models

What condition must hold on the severity distribution for Panjer's recursion to apply?

Answer

The severity distribution must be discretized onto a grid of possible claim sizes $0, h, 2h, 3h, \dots$ with a fixed span h .

#99 · Probability Models

What is g_0 in Panjer's recursion?

Answer

$g_0 = P_N(f_0)$, the frequency distribution's probability generating function evaluated at f_0 , the probability that severity equals zero.

#100 · Probability Models

For a $\text{Poisson}(\lambda)$ frequency, what does g_0 equal in Panjer's recursion?

Answer

$g_0 = \exp(\lambda(f_0 - 1))$

#101 · Probability Models

What is the general form of Panjer's recursion for g_k when $k \geq 1$?

Answer

$$g_k = [1 \div (1 - a \times f_0)] \times \sum_{j=1}^k (a + b \times j \div k) \times f_j \times g_{(k-j)}$$

#102 · Probability Models

What computational advantage does Panjer's recursion have over direct convolution?

Answer

It builds the aggregate distribution with a single forward pass at cost proportional to k^2 , instead of computing an infinite mixture of n -fold convolutions weighted by $P(N=n)$.

#103 · Probability Models

In direct convolution (without Panjer's recursion), how would you compute the distribution of $X_1+X_2+X_3$?

Answer

By convolving the severity distribution with itself twice: first compute the 2-fold convolution for X_1+X_2 , then convolve that result again with the severity distribution for the 3-fold convolution.

#104 · Probability Models

For Panjer's recursion with a Poisson frequency ($a=0$), what does the general recursion simplify to?

Answer

$g_k = b \times \sum_{j=1}^k (j \div k) \times f_j \times g_{k-j}$, since $a=0$ removes the constant term and the $1 \div (1-a \times f_0)$ factor becomes 1.

#105 · Probability Models

With $N \sim \text{Poisson}(\lambda=2)$ and $f_0=0$, $f_1=0.6$, $f_2=0.3$, $f_3=0.1$, what is g_0 ?

Answer

$$g_0 = \exp(2(0-1)) = \exp(-2) = 0.1353$$

#106 · Probability Models

With $N \sim \text{Poisson}(\lambda=2)$, $b=2$, $f_1=0.6$, and $g_0=0.1353$, what is g_1 ?

Answer

$$g_1 = b \times (1 \div 1) \times f_1 \times g_0 = 2 \times 1 \times 0.6 \times 0.1353 = 0.1624$$

#107 · Probability Models

With $b=2$, $f_1=0.6$, $f_2=0.3$, $g_0=0.1353$, $g_1=0.1624$, what is g_2 using Panjer's recursion?

Answer

$$g_2 = 2 \times [(0.5)(0.6)(0.1624) + (1)(0.3)(0.1353)] = 2 \times [0.04872 + 0.04059] = 2 \times 0.08931 = 0.1786$$

#108 · Probability Models

What is a discretized severity distribution and why is it needed for Panjer's recursion?

Answer

It is the severity distribution rounded or mapped onto a fixed grid of values $(0, h, 2h, \dots)$, needed because Panjer's recursion is defined for discrete probability mass functions.

#109 · Probability Models

What is stop-loss insurance?

Answer

A reinsurance arrangement that pays the excess of the aggregate loss S over a fixed retention d , so the reinsurer pays $(S-d)_+ = \max(S-d, 0)$.

#110 · Probability Models

What is the formula for the net stop-loss premium at retention d for a discrete aggregate distribution?

Answer

$E[(S-d)_+] = \sum_{s > d} (s-d) \times P(S=s)$

#111 · Probability Models

What is the survival-function form of the stop-loss premium formula?

Answer

$$E[(S-d)_+] = \int_d^{\infty} [1 - F_S(x)] dx$$

#112 · Probability Models

Why are Panjer's recursion and stop-loss premiums almost always tested together?

Answer

Because the full aggregate distribution built through Panjer's recursion is typically the direct input needed to compute a stop-loss premium at any retention level.

#113 · Probability Models

What does it mean for a retention d to be applied to an aggregate loss S in a stop-loss treaty?

Answer

The reinsurer only pays the amount by which S exceeds d ; if S is at or below d , the reinsurer pays nothing.

#114 · Probability Models

For a binomial(m, q) frequency, why is a always negative?

Answer

Because $a = -q/(1-q)$, and since q and $1-q$ are both positive for $0 < q < 1$, the ratio $q/(1-q)$ is positive, making a strictly negative.

#115 · Probability Models

Why does the negative binomial distribution always have b less than a when $r < 1$?

Answer

Because $b = (r-1) \times \text{beta} \div (1+\text{beta})$; when $r < 1$, $(r-1)$ is negative, making b negative while $a = \text{beta} \div (1+\text{beta})$ stays positive.

#116 · Probability Models

What real-world consequence follows from underpricing overdispersion in a claim count model?

Answer

Using a Poisson model when the true process is overdispersed understates the variability of aggregate losses, which can lead to underpriced tail risk in reinsurance or inadequate reserve margins.

#117 · Probability Models

In the Poisson-gamma mixture interpretation of the negative binomial, what does the gamma distribution represent?

Answer

The gamma distribution represents uncertainty or heterogeneity in the true claim rate λ across policyholders, before conditioning on a specific policyholder's Poisson claim process.

#118 · Probability Models

What happens to the compound-variance formula if the frequency N is degenerate (always equal to a fixed constant n)?

Answer

$\text{Var}(N) = 0$, so $\text{Var}(S)$ reduces to just $n \times \text{Var}(X)$, since the frequency contributes no variability.

#119 · Probability Models

Why must the severity random variables X_i be independent of the frequency N in the collective risk model?

Answer

So that the number of claims does not influence or correlate with the size of each individual claim, which keeps the compound-variance formula and Panjer's recursion valid.

#120 · Probability Models

What is the practical effect of a fixed maximum m in a binomial frequency model on the aggregate loss distribution?

Answer

It caps the maximum possible aggregate loss at m times the maximum possible severity, since no more than m claims can ever occur.

#121 · Probability Models

How would you verify from sample data whether to use a Poisson or negative binomial frequency model?

Answer

Compare the sample mean and sample variance of claim counts; if the sample variance is noticeably larger than the sample mean, this indicates overdispersion and favors the negative binomial.

#122 · Probability Models

What is the relationship between p_1 and p_0 under the $(a,b,0)$ recursion?

Answer

$p_1 = p_0 \times (a + b)$, since the recursion $p_k \div p_{(k-1)} = a + b \div k$ gives $p_1 \div p_0 = a + b \div 1 = a + b$.

#123 · Probability Models

Does Panjer's recursion require knowing the full frequency pmf in advance, or only a and b?

Answer

Only a , b , and p_0 (equivalently g_0) are needed; the recursion generates every subsequent probability without needing the full frequency pmf listed out in advance.

#124 · Probability Models

Why is convolution of discrete severity distributions computationally expensive compared to Panjer's recursion?

Answer

Direct convolution requires computing a separate n -fold convolution of the severity distribution for every possible value of N and then weighting and summing over all n , which grows rapidly in cost, while Panjer's recursion builds the result in a single efficient forward pass.

#125 · Probability Models

If lambda increases for a Poisson frequency while severity stays fixed, what happens to both $E[S]$ and $\text{Var}(S)$?

Answer

Both $E[S] = \lambda \times E[X]$ and the $\text{Var}(N)$ term $\lambda \times (E[X])^2$ in $\text{Var}(S)$ increase directly with λ , so both the expected aggregate loss and its variance rise.

#126 · Statistical Inference

What is the null hypothesis H_0 ?

Answer

The default claim of no effect or no difference, assumed true until the sample data provide strong evidence against it.

#127 · Statistical Inference

What is the alternative hypothesis H_1 ?

Answer

The claim the analyst suspects may be true instead of H_0 , which is adopted only if the data provide sufficient evidence against H_0 .

#128 · Statistical Inference

Define a Type I error.

Answer

Rejecting H_0 when H_0 is actually true, a false positive. Its probability is the significance level, α .

#129 · Statistical Inference

Define a Type II error.

Answer

Failing to reject H_0 when H_0 is actually false, a false negative. Its probability is denoted β .

#130 · Statistical Inference

What is the power of a hypothesis test?

Answer

1 minus β , the probability of correctly rejecting a false H_0 .

#131 · Statistical Inference

What happens to power as sample size increases, holding alpha fixed?

Answer

Power increases, because a larger sample reduces sampling variability and makes a true effect easier to detect.

#132 · Statistical Inference

What is the tradeoff between alpha and beta at a fixed sample size?

Answer

Lowering alpha to reduce Type I error raises beta (and lowers power); the two error probabilities move in opposite directions for a fixed n .

#133 · Statistical Inference

Can a hypothesis test prove H_0 is true?

Answer

No. Failing to reject H_0 only means insufficient evidence was found against it, not that H_0 has been proven true.

#134 · Statistical Inference

What is the p-value, precisely defined?

Answer

The probability, assuming H_0 is true, of observing a test statistic at least as extreme as the one actually observed.

#135 · Statistical Inference

What decision rule uses the p-value?

Answer

Reject H_0 if the p-value is less than or equal to α ; otherwise fail to reject H_0 .

#136 · Statistical Inference

Is the p-value the probability that H_0 is true?

Answer

No, this is a common misinterpretation. The p-value is a conditional probability computed assuming H_0 is true, not the probability that H_0 itself is true.

#137 · Statistical Inference

Why can a trivially small effect still produce a very small p-value?

Answer

Because the p-value reflects both effect size and sample size; with a large enough sample, even a practically meaningless difference becomes statistically significant.

#138 · Statistical Inference

How does a two-sided test at level alpha split the rejection region?

Answer

Into two tails, each with probability alpha divided by 2.

#139 · Statistical Inference

One-sample z-test statistic formula (sigma known)?

Answer

$$z = (\bar{x} - \mu_0) \div (\sigma \div \sqrt{n}).$$

#140 · Statistical Inference

When should a t-test be used instead of a z-test for a mean?

Answer

When the population standard deviation σ is unknown and the sample size is small (commonly n less than 30), so the sample standard deviation s must be used.

#141 · Statistical Inference

One-sample t-test statistic formula?

Answer

$t = (\bar{x} - \mu_0) \div (s \div \sqrt{n})$, with $n - 1$ degrees of freedom.

#142 · Statistical Inference

Why does the t distribution have heavier tails than the normal distribution?

Answer

Because it accounts for the extra uncertainty of estimating sigma from the sample using s, which the z distribution does not need to do.

#143 · Statistical Inference

What happens to the t distribution as sample size n grows large?

Answer

It converges to the standard normal distribution, so the t-test and z-test become numerically indistinguishable.

#144 · Statistical Inference

$z(0.025)$, the two-sided critical value at $\alpha = 0.05$?

Answer

1.960.

#145 · Statistical Inference

$z(0.005)$, the two-sided critical value at $\alpha = 0.01$?

Answer

2.576.

#146 · Statistical Inference

Two-sample z-test statistic for a difference in means, variances known?

Answer

$$z = (\bar{x}_1 - \bar{x}_2) \div \sqrt{(\sigma_1^2 \div n_1 + \sigma_2^2 \div n_2)}.$$

#147 · Statistical Inference

Pooled variance formula for a two-sample t-test with equal population variances assumed?

Answer

$$s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] \div (n_1 + n_2 - 2).$$

#148 · Statistical Inference

Pooled two-sample t-statistic formula?

Answer

$$t = (\bar{x}_1 - \bar{x}_2) \div \sqrt{s_p^2 (1/n_1 + 1/n_2)}, \text{ with } n_1 + n_2 - 2 \text{ degrees of freedom.}$$

#149 · Statistical Inference

What is Welch's approximation used for?

Answer

Comparing two sample means when population variances cannot be assumed equal; it uses the unpooled standard error and an adjusted, generally non-integer, degrees-of-freedom formula.

#150 · Statistical Inference

What does a 95% confidence interval actually mean?

Answer

Across repeated sampling, 95% of intervals built this way would capture the true parameter. It is NOT a 95% probability that the true value lies in this one already-computed interval.

#151 · Statistical Inference

Confidence interval for a mean when sigma is known?

Answer

$$\bar{x} \pm z(\alpha/2) \times \sigma / \sqrt{n}.$$

#152 · Statistical Inference

Confidence interval for a mean when sigma is unknown?

Answer

$$\bar{x} \pm t(\alpha/2, n-1) \times s / \sqrt{n}.$$

#153 · Statistical Inference

Confidence interval for a population variance uses which distribution?

Answer

The chi-square distribution with $n - 1$ degrees of freedom. The resulting interval is not symmetric around the sample variance.

#154 · Statistical Inference

Confidence interval formula for a population variance?

Answer

$$\left[(n-1)s^2 \div \text{chi-square}(\alpha \div 2, n-1), (n-1)s^2 \div \text{chi-square}(1-\alpha \div 2, n-1) \right].$$

#155 · Statistical Inference

In the variance confidence interval, which chi-square critical value goes in the denominator of the LOWER bound?

Answer

The UPPER chi-square critical value, $\chi^2(\alpha \div 2, n-1)$; this reversal is a common source of exam errors.

#156 · Statistical Inference

Confidence interval for a population proportion p?

Answer

$\hat{p} \pm z(\alpha \div 2) \times \sqrt{\hat{p}(1-\hat{p}) \div n}$, valid when $n \times \hat{p}$ and $n \times (1-\hat{p})$ are both reasonably large.

#157 · Statistical Inference

How are a two-sided hypothesis test and a confidence interval at the same alpha connected?

Answer

$H_0: \mu = \mu_0$ is rejected at level α if and only if μ_0 falls outside the corresponding $(1-\alpha)$ confidence interval for μ .

#158 · Statistical Inference

What does the chi-square goodness-of-fit test statistic measure?

Answer

The discrepancy between observed counts O_i and expected counts E_i across k classes: $\text{chi-square} = \text{sum of } (O_i - E_i)^2 \div E_i$.

#159 · Statistical Inference

Degrees of freedom for the chi-square goodness-of-fit test?

Answer

$k - 1 - r$, where k is the number of classes and r is the number of parameters estimated from the same data used to build the classes.

#160 · Statistical Inference

What rule of thumb governs class sizes in a chi-square goodness-of-fit test?

Answer

Each expected count E_i should generally be at least 5; adjacent classes with smaller expected counts are often combined.

#161 · Statistical Inference

What is a key weakness of the chi-square goodness-of-fit test?

Answer

It requires grouping continuous data into discrete classes, which discards information and makes results sensitive to the arbitrary choice of class boundaries.

#162 · Statistical Inference

What does the Kolmogorov-Smirnov (K-S) test statistic D measure?

Answer

The supremum (maximum) vertical distance between the empirical CDF $F_n(x)$ and the fitted CDF $F(x)$: $D = \sup \text{ over } x \text{ of } |F_n(x) - F(x)|$.

#163 · Statistical Inference

Why can the K-S statistic D be computed exactly without checking every real number x ?

Answer

Because the empirical CDF $F_n(x)$ is a step function that only changes value at the observed data points, so D only needs to be checked just before and after each order statistic.

#164 · Statistical Inference

What is a key limitation of the K-S test for actuarial severity modeling?

Answer

It weights every part of the distribution equally, so it can miss a poor fit in the tail even if the fit is good in the center, which matters for reserving and pricing accuracy.

#165 · Statistical Inference

What is the Anderson-Darling test statistic designed to emphasize?

Answer

Discrepancies between the empirical and fitted CDFs in the tails of the distribution, making it more tail-sensitive than the K-S test.

#166 · Statistical Inference

What is the weight function in the Anderson-Darling statistic and why does it matter?

Answer

$1 \div (F(x)(1-F(x)))$; it grows very large as $F(x)$ approaches 0 or 1, giving much more weight to gaps in the extreme tails than to gaps near the median.

#167 · Statistical Inference

Why is the Anderson-Darling test generally preferred over K-S for severity distributions?

Answer

Because tail behavior drives reserving and pricing accuracy, and A-D is specifically weighted to be sensitive to tail discrepancies, while K-S treats the tails and center equally.

#168 · Statistical Inference

What does it mean for two models to be nested?

Answer

The simpler model is a special case of the more complex model, obtained by restricting one or more parameters, often by setting them to a fixed value such as zero.

#169 · Statistical Inference

Likelihood ratio test (LRT) statistic formula?

Answer

$LRT = 2 \times (\ln L1 - \ln L0) = 2 \times \ln(L1 \div L0)$, where $L1$ is the maximized likelihood of the full model and $L0$ is the maximized likelihood of the restricted model.

#170 · Statistical Inference

What distribution does the LRT statistic follow under H_0 , and with how many degrees of freedom?

Answer

Approximately a chi-square distribution, with degrees of freedom equal to the number of additional parameters in the full model relative to the restricted model.

#171 · Statistical Inference

Why is L_1 always at least as large as L_0 in a likelihood ratio test?

Answer

Because the restricted model is a special case of the full model, so the full model has at least as much flexibility and can never fit worse.

#172 · Statistical Inference

Give an actuarial example of nested severity distributions suited to an LRT.

Answer

A two-parameter Weibull versus a one-parameter exponential; a Weibull with its shape parameter fixed at 1 reduces exactly to an exponential.

#173 · Statistical Inference

In Worked Example 1, a sample of $n = 16$ claims gives $\bar{x} = 4550$, $s = 620$, testing against $\mu_0 = 4200$. What is the t-statistic?

Answer

$$t = (4550 - 4200) \div (620 \div \sqrt{16}) = 350 \div 155 = 2.258.$$

#174 · Statistical Inference

In Worked Example 1, with $t = 2.258$ and critical value $t(0.025, 15) = 2.131$, what is the conclusion at $\alpha = 0.05$?

Answer

Reject H_0 ; there is statistically significant evidence the mean severity differs from the 4,200 benchmark, since $|t| = 2.258$ exceeds 2.131.

#175 · Statistical Inference

In Worked Example 1, would the same $t = 2.258$ lead to rejection at $\alpha = 0.01$?

Answer

No; the critical value $t(0.005, 15) = 2.947$ exceeds 2.258, so H_0 would not be rejected at the 1% level.

#176 · Statistical Inference

In Worked Example 2, with observed counts 40, 55, 50, 35, 20 and expected counts 36, 60, 52, 32, 20, what is the chi-square statistic?

Answer

Approximately 1.219, computed as the sum of $(O_i - E_i)^2 \div E_i$ across the five bands.

#177 · Statistical Inference

In Worked Example 2, with $r = 0$ parameters estimated and $k = 5$ classes, how many degrees of freedom does the chi-square test use?

Answer

$k - 1 - r = 5 - 1 - 0 = 4$ degrees of freedom.

#178 · Statistical Inference

In Worked Example 2, with chi-square = 1.219 and critical value chi-square(0.05, 4) = 9.488, what is the conclusion?

Answer

Fail to reject H_0 ; the observed statistic is far below the critical value, so there is no significant evidence of a poor fit.

#179 · Statistical Inference

What is a rejection region (critical region)?

Answer

The set of test statistic values that leads to rejection of H_0 , determined by the chosen alpha and the reference distribution.

#180 · Statistical Inference

Difference between a one-sided and a two-sided hypothesis test?

Answer

A one-sided test places the entire alpha in a single tail (testing greater than or less than); a two-sided test splits alpha into two tails (testing simply not equal to).

#181 · Statistical Inference

Why is statistical significance not the same as practical significance?

Answer

A statistically significant p-value can arise from a trivially small effect measured with a very large sample; the actuary must separately assess whether the effect size is materially important to the business.

#182 · Statistical Inference

When comparing two independent population means, what is tested under H_0 ?

Answer

$H_0: \mu_1 = \mu_2$, that the two population means are equal, tested using the difference $\bar{x}_1$ minus $\bar{x}_2$.

#183 · Statistical Inference

What caveat applies to K-S critical value tables when distribution parameters are estimated from the same sample?

Answer

The standard K-S critical values are technically not valid in that case; adjusted critical values or simulation are needed instead.

#184 · Statistical Inference

What general actuarial modeling principle does the LRT support?

Answer

The principle of parsimony: prefer the simpler model unless the added complexity of the full model produces a statistically significant improvement in fit.

#185 · Statistical Inference

How does increasing alpha affect power, holding sample size fixed?

Answer

Increasing alpha raises power (a looser rejection threshold catches more true effects), but also increases the Type I error rate.

#186 · Statistical Inference

What is the empirical distribution function $F_n(x)$?

Answer

The proportion of sample observations less than or equal to x , built directly from the n data points; it is a step function.

#187 · Statistical Inference

Is the p-value a measure of effect size?

Answer

No. The p-value reflects both effect size and sample size together; it does not by itself measure how large or important an effect is.

#188 · Statistical Inference

What must an actuary check before applying the chi-square confidence interval for a variance or the normal approximation for a proportion?

Answer

For the proportion interval, that $n \times \hat{p}$ and $n \times (1 - \hat{p})$ are both reasonably large (commonly at least 5 or 10); the chi-square variance interval relies on the data being approximately normal.

#189 · Statistical Inference

What does Bayes' theorem state in terms of prior, likelihood, and posterior?

Answer

The posterior distribution $\pi(\theta \text{ given } x)$ equals the likelihood $f(x \text{ given } \theta)$ times the prior $\pi(\theta)$, divided by the marginal probability of the data $f(x)$. In shorthand: posterior is proportional to likelihood times prior.

#190 · Statistical Inference

What is the prior distribution in Bayesian statistics?

Answer

The probability distribution $\pi(\theta)$ that reflects belief about an unknown parameter θ before any new data is observed, often based on industry experience or expert judgment.

#191 · Statistical Inference

What role does the marginal density $f(x)$ play in Bayes' theorem?

Answer

It is the normalizing constant, obtained by integrating the likelihood times prior over all values of θ , that makes the posterior density integrate to 1. It does not depend on θ .

#192 · Statistical Inference

What is a conjugate prior?

Answer

A prior distribution that, when combined with a given likelihood, produces a posterior distribution in the same distributional family as the prior.

#193 · Statistical Inference

In the gamma-Poisson conjugate pair, what is the prior distribution and what is the likelihood?

Answer

The prior on the Poisson mean θ is $\text{gamma}(\alpha, \beta)$. The likelihood, given n observed Poisson counts, is proportional to θ raised to the sum of counts, times e to the negative n times θ .

#194 · Statistical Inference

Give the gamma-Poisson posterior updating formulas.

Answer

$\alpha_{\text{posterior}} = \alpha_{\text{prior}} + \text{sum of observed claim counts}$. $\beta_{\text{posterior}} = \beta_{\text{prior}} + n$, where n is the number of exposure periods observed.

#195 · Statistical Inference

If the prior is $\text{gamma}(\alpha=4, \beta=2)$ and 6 claims are observed over 3 exposure years, what is the posterior distribution?

Answer

$\text{gamma}(\alpha_{\text{posterior}} = 4+6 = 10, \beta_{\text{posterior}} = 2+3 = 5)$, with posterior mean $10 \div 5 = 2.0$.

#196 · Statistical Inference

What is the posterior mean of a $\text{gamma}(\alpha, \beta)$ distribution?

Answer

α divided by β .

#197 · Statistical Inference

In the beta-binomial conjugate pair, what is the prior distribution?

Answer

A beta distribution with parameters a and b , used as the prior on the binomial success probability θ , on the interval $(0,1)$.

#198 · Statistical Inference

Give the beta-binomial posterior updating formulas.

Answer

$a_{\text{posterior}} = a_{\text{prior}} + x$ (observed successes).
 $b_{\text{posterior}} = b_{\text{prior}} + (m \text{ minus } x)$ (observed failures), where m is the number of trials.

#199 · Statistical Inference

What is the posterior mean of a beta(a, b) distribution?

Answer

a divided by (a + b).

#200 · Statistical Inference

In the normal-normal conjugate pair, what distributions are assumed for the data and the prior?

Answer

The data X given θ is normal with mean θ and known variance σ^2 . The prior on θ is normal with mean μ_0 and variance τ^2 .

#201 · Statistical Inference

What is the posterior mean formula in the normal-normal conjugate pair?

Answer

posterior mean = Z times $\bar{x}$ + (1 minus Z) times μ_0 , where $Z = (n\div\sigma^2)$ divided by $[(n\div\sigma^2) + (1\div\tau^2)]$.

#202 · Statistical Inference

What is the posterior variance formula in the normal-normal conjugate pair?

Answer

1 divided by $[(n\div\sigma^2) + (1\div\tau^2)]$, which is always smaller than either the prior variance τ^2 or $\sigma^2\div n$ alone.

#203 · Statistical Inference

How does the Bayesian posterior mean relate to actuarial credibility theory?

Answer

In conjugate models, the posterior mean can be rewritten as Z times the observed data mean plus $(1 - Z)$ times the prior (manual) mean, exactly matching the credibility-weighted-average formula.

#204 · Statistical Inference

In the gamma-Poisson posterior mean, what is the credibility factor Z equal to?

Answer

$Z = n \div (beta_prior + n)$, where n is the number of exposure periods observed and $beta_prior$ is the prior gamma rate parameter.

#205 · Statistical Inference

What happens to the credibility factor Z as the number of observed exposure periods n grows very large?

Answer

Z approaches 1, meaning the posterior estimate relies almost entirely on the observed data and gives almost no weight to the prior mean.

#206 · Statistical Inference

What happens to the credibility factor Z when the prior is very concentrated (low prior variance, high confidence)?

Answer

Z shrinks toward 0, so the posterior estimate leans heavily on the prior mean and gives little weight to newly observed data.

#207 · Statistical Inference

What is the posterior mode, also called the MAP estimate?

Answer

The value of θ that maximizes the posterior density, i.e. the maximum a posteriori estimate.

#208 · Statistical Inference

What loss function does the posterior mean minimize?

Answer

Expected squared error loss.

#209 · Statistical Inference

What loss function does the posterior median minimize?

Answer

Expected absolute error loss.

#210 · Statistical Inference

For a gamma(alpha, beta) posterior with alpha greater than 1, what is the posterior mode?

Answer

(alpha minus 1) divided by beta, which always differs from the posterior mean of alpha divided by beta since the mode subtracts 1 from alpha in the numerator.

#211 · Statistical Inference

When do the posterior mean, median, and mode coincide?

Answer

When the posterior distribution is symmetric and unimodal, such as in the normal-normal conjugate case.

#212 · Statistical Inference

What is a Bayesian credible interval?

Answer

An interval $[L, U]$ such that the posterior probability θ falls between L and U equals a stated level, for example the integral of the posterior density from L to U equals 0.90 for a 90 percent credible interval.

#213 · Statistical Inference

In a frequentist confidence interval, is theta treated as random or fixed?

Answer

Theta is treated as a fixed, unknown constant. The interval itself, built from a random sample, is the random quantity.

#214 · Statistical Inference

What is the correct interpretation of a 90 percent frequentist confidence interval?

Answer

If the sampling procedure were repeated many times, 90 percent of the resulting intervals would contain the true fixed parameter theta. It is not a direct probability statement about one already-computed interval.

#215 · Statistical Inference

In a Bayesian credible interval, is theta treated as random or fixed?

Answer

Theta is treated as the random variable, and once the posterior is computed, the credible interval is a fixed numeric interval with a direct probability statement attached to it.

#216 · Statistical Inference

What is the key conceptual difference between a Bayesian credible interval and a frequentist confidence interval?

Answer

A credible interval allows a direct probability statement about theta given the observed data, since theta is random in the Bayesian framework; a confidence interval only supports a statement about long-run coverage across repeated sampling, since theta is fixed in the frequentist framework.

#217 · Statistical Inference

What is an equal-tailed credible interval?

Answer

A credible interval that places the stated significance level evenly in each tail of the posterior, for example the 5th and 95th posterior percentiles for a 90 percent interval.

#218 · Statistical Inference

What is a highest posterior density (HPD) interval?

Answer

The shortest possible interval containing a stated amount of posterior probability mass. For a skewed posterior it is narrower than the equal-tailed interval, and its tail probabilities need not be equal.

#219 · Statistical Inference

What is Monte Carlo simulation used for?

Answer

Estimating a quantity of interest, such as an expected value, probability, or quantile, by generating many pseudo-random draws from a probability model and computing the corresponding sample statistic.

#220 · Statistical Inference

According to the law of large numbers, what does the average of many simulated draws converge to?

Answer

The true mean μ of the distribution being simulated from, as the number of simulated trials n grows large.

#221 · Statistical Inference

How does the standard error of a Monte Carlo mean estimate behave as the number of simulated trials n increases?

Answer

It decreases proportionally to 1 divided by the square root of n , so quadrupling the number of trials roughly halves the simulation error.

#222 · Statistical Inference

How is a quantile, such as the 99.5th percentile, estimated from simulated data?

Answer

Sort the simulated values from smallest to largest and read off the value at the rank corresponding to that percentile (approximately p times n , possibly interpolated).

#223 · Statistical Inference

Why is Monte Carlo simulation especially useful for aggregate loss models with policy limits and deductibles?

Answer

Because a closed-form formula for the expected value or quantile of the resulting distribution is often difficult or impossible to derive analytically, while simulating many years of claim experience and applying policy terms directly is straightforward.

#224 · Statistical Inference

What is the inverse transform method used for?

Answer

Simulating a random draw from a distribution whose cumulative distribution function (CDF) is known and invertible, using a uniform random number as the starting point.

#225 · Statistical Inference

State the inverse transform method's core mathematical fact.

Answer

If U is Uniform(0,1) and F is a continuous, strictly increasing CDF, then $X = F^{-1}(U)$ has exactly the distribution described by F .

#226 · Statistical Inference

List the three steps of the inverse transform method.

Answer

1. Generate a Uniform(0,1) draw u .
2. Set $x = F^{-1}(u)$.
3. Repeat independently for each additional simulated draw needed.

#227 · Statistical Inference

What is the inverse CDF of an Exponential distribution with mean θ ?

Answer

$x = -\theta \ln(1 - u)$, derived by setting $F(x) = 1 - e^{-x/\theta}$ equal to u and solving for x .

#228 · Statistical Inference

Using the inverse transform method, simulate an Exponential claim size with mean 500 given a uniform draw of $u = 0.70$.

Answer

$x = -500 \ln(1 - 0.70) = -500 \ln(0.30)$, which is approximately $-500 \times (-1.2040) = 602.0$.

#229 · Statistical Inference

In the inverse transform method applied to an Exponential distribution, what happens when the uniform draw u is close to 1?

Answer

The simulated value becomes very large, since $\ln(1 - u)$ approaches negative ∞ , consistent with the long right tail of the exponential distribution.

#230 · Statistical Inference

How is the inverse transform method adapted for discrete distributions?

Answer

The interval $(0,1)$ is partitioned into consecutive sub-intervals whose lengths equal the probability mass at each outcome, and whichever outcome's sub-interval contains the generated uniform draw is selected.

#231 · Statistical Inference

What alternative simulation method is used when a distribution's CDF cannot be inverted in closed form?

Answer

Numerical root-finding or algorithms such as the accept-reject method.

#232 · Statistical Inference

What is the bootstrap method used for?

Answer

Estimating the sampling distribution of a statistic, such as a sample mean or a fitted model parameter, by resampling from the observed data, when the population distribution is unknown or a closed-form formula for the statistic's variability is unavailable.

#233 · Statistical Inference

What distinguishes the nonparametric bootstrap's resampling procedure?

Answer

Each bootstrap sample is drawn from the original n observations **WITH** replacement, so some original observations appear multiple times and others do not appear at all in a given bootstrap sample.

#234 · Statistical Inference

List the four steps of the nonparametric bootstrap procedure.

Answer

1. Draw a bootstrap sample of size n with replacement from the original data.
2. Compute the statistic of interest on that sample.
3. Repeat many times (B replications).
4. Use the empirical distribution of the B replications to approximate the true sampling distribution.

#235 · Statistical Inference

How is the standard error of a statistic estimated using the bootstrap?

Answer

By computing the standard deviation of the B bootstrap replications of that statistic.

#236 · Statistical Inference

What is the percentile method for constructing a bootstrap confidence interval?

Answer

Sort the B bootstrap replications of the statistic and take the $\alpha \div 2$ and $(1 \text{ minus } \alpha \div 2)$ empirical percentiles directly as the lower and upper confidence bounds.

#237 · Statistical Inference

For a 90 percent bootstrap confidence interval with $B = 1,000$ replications using the percentile method, which ranked values form the bounds?

Answer

The 50th smallest value (5th percentile) as the lower bound and the 950th smallest value (95th percentile) as the upper bound.

#238 · Statistical Inference

Given claim sizes 200, 350, 400, 600, 950, what is the original sample mean?

Answer

$(200+350+400+600+950) \div 5 = 2500 \div 5 = 500.$

#239 · Statistical Inference

Why might an actuary prefer the bootstrap over a large-sample asymptotic normality approximation for a fitted GLM parameter?

Answer

Asymptotic approximations can be inaccurate in small samples; the bootstrap empirically characterizes the estimator's variability directly from resampled data without relying on that large-sample assumption.

#240 · Statistical Inference

What is the main computational tradeoff of bootstrap methods?

Answer

They require many repeated model fits or statistic calculations (often thousands), which can be demanding when the underlying statistic is expensive to compute.

#241 · Statistical Inference

What is the parametric bootstrap, as opposed to the nonparametric bootstrap?

Answer

A bootstrap variant that resamples from a fitted parametric model itself rather than resampling directly from the original observed data.

#242 · Statistical Inference

In the gamma-Poisson pair, what does the prior mean α/β represent in credibility terms?

Answer

It is the manual or a priori mean, the estimate used before any experience data is incorporated.

#243 · Statistical Inference

Why is the posterior distribution only proportional to likelihood times prior, rather than exactly equal to it?

Answer

Because the normalizing constant $f(x)$ has been dropped; it is recovered afterward as whatever value makes the posterior density integrate to 1 over θ .

#244 · Statistical Inference

What role does the shape of the posterior, as a function of θ , play in solving conjugate prior problems on the exam?

Answer

Recognizing the posterior's functional shape in θ is usually enough to identify which known distribution family it belongs to and read off its parameters, avoiding the need to compute the full normalizing integral.

#245 · Statistical Inference

How does a very diffuse (high variance) prior affect the posterior compared to a very concentrated (low variance) prior?

Answer

A diffuse prior lets the observed data dominate the posterior almost immediately, while a concentrated prior keeps the posterior close to the prior mean even after a moderate amount of data is observed.

#246 · Statistical Inference

In the beta-binomial pair, what does the parameter m represent?

Answer

The number of binomial trials observed, such as the number of policyholders or exposure units examined.

#247 · Statistical Inference

What is the defining feature that makes a resampling method 'nonparametric' in the bootstrap context?

Answer

It draws resamples directly from the empirical observed data without assuming the data follows any specific parametric distribution family.

#248 · Statistical Inference

If a second uniform draw of $u = 0.20$ is used with the `Exponential(mean=500)` inverse transform formula, what is the simulated value?

Answer

$x = \text{negative } 500 \text{ times } \ln(1 \text{ minus } 0.20) = \text{negative } 500 \text{ times } \ln(0.80)$, approximately negative 500 times (negative 0.2231) = 111.6.

#249 · Statistical Inference

Why does the exponential inverse transform formula produce small simulated values for small uniform draws?

Answer

When u is close to 0, $\ln(1 - u)$ is close to 0, so $x = -\theta \ln(1 - u)$ is close to 0, matching the exponential distribution's concentration of probability near zero.

#250 · Statistical Inference

What general pattern do all three conjugate posterior means (gamma-Poisson, beta-binomial, normal-normal) share?

Answer

Each posterior mean is a weighted average of the observed sample statistic and the prior mean, with the weight on the data increasing in sample size and decreasing in prior confidence (or increasing in prior variance).

#251 · Generalized Linear Models

Why is OLS regression a poor fit for claim severity data?

Answer

Severity is non-negative and right skewed, but a normal distribution allows negative values and is symmetric, so OLS can produce nonsensical negative predictions and misrepresents the shape of the data.

#252 · Generalized Linear Models

Why is OLS regression a poor fit for claim count data?

Answer

Claim counts are non-negative integers, but OLS predicts a continuous response that can be fractional or negative, which does not match how claim counts actually behave.

#253 · Generalized Linear Models

What is heteroscedasticity, and why does it matter for insurance data?

Answer

Heteroscedasticity means the variance of the response changes with the mean. Insurance data typically show variance growing as the mean grows, which violates the OLS assumption of constant variance and leads to incorrect standard errors.

#254 · Generalized Linear Models

Write the general form of an exponential family distribution.

Answer

$f(y; \theta, \phi) = \exp[(y \times \theta - b(\theta)) \div a(\phi) + c(y, \phi)]$, where θ is the canonical parameter, ϕ is the dispersion parameter, and b and c are functions defining the distribution.

#255 · Generalized Linear Models

In the exponential family form, what does theta represent?

Answer

Theta is the canonical parameter, also called the natural parameter, and it is the parameter through which the mean of the distribution is expressed.

#256 · Generalized Linear Models

In the exponential family form, what does phi represent?

Answer

Phi is the dispersion parameter, which controls the scale of the variance of the distribution, separate from the canonical parameter theta.

#257 · Generalized Linear Models

In exponential family notation, how is the mean of Y expressed in terms of $b(\theta)$?

Answer

$E(Y) = \mu = b'(\theta)$, the first derivative of b with respect to θ .

#258 · Generalized Linear Models

In exponential family notation, how is the variance of Y expressed in terms of $b(\theta)$?

Answer

$\text{Var}(Y) = a(\phi) \times b''(\theta)$, the dispersion function times the second derivative of b with respect to θ .

#259 · Generalized Linear Models

What is the variance function $V(\mu)$?

Answer

The variance function is $b''(\theta)$ re-expressed as a function of μ (since $\mu = b'(\theta)$). It captures how the variance of Y depends on the mean for a given distribution.

#260 · Generalized Linear Models

Write the general mean-variance relationship used in GLMs.

Answer

$\text{Var}(Y) = \phi \times V(\mu)$, where ϕ is the dispersion parameter and $V(\mu)$ is the variance function determined by the chosen distribution.

#261 · Generalized Linear Models

What is the variance function for the normal distribution?

Answer

$V(\mu) = 1$, a constant, meaning the variance does not depend on the mean, matching the homoscedasticity assumption of OLS.

#262 · Generalized Linear Models

What is the variance function for the Poisson distribution?

Answer

$V(\mu) = \mu$, so the variance equals the mean, which is why $\text{Var}(Y) = \mu$ for a Poisson variable with dispersion parameter fixed at 1.

#263 · Generalized Linear Models

What is the variance function for the gamma distribution?

Answer

$V(\mu) = \mu^2$, so the variance grows with the square of the mean, and the coefficient of variation stays constant regardless of the size of the mean.

#264 · Generalized Linear Models

What is the variance function for the binomial distribution?

Answer

$V(\mu) = \mu(1-\mu)$, where μ is between 0 and 1. This is largest at $\mu = 0.5$ and shrinks toward zero as μ approaches 0 or 1.

#265 • Generalized Linear Models

What does it mean for a claim count variable to be overdispersed?

Answer

Overdispersion means the sample variance exceeds the sample mean, which contradicts the Poisson assumption that variance equals mean, and often signals that a negative binomial model would fit better.

#266 • Generalized Linear Models

Name the three components of a generalized linear model.

Answer

The random component (the distribution of the response), the systematic component (the linear predictor), and the link function (which connects the mean to the linear predictor).

#267 · Generalized Linear Models

What is the random component of a GLM?

Answer

The random component is the choice of exponential family distribution assumed for the response variable Y , such as Poisson, gamma, binomial, or normal.

#268 · Generalized Linear Models

What is the systematic component of a GLM?

Answer

The systematic component, or linear predictor, is $\eta = X \times \beta$, a linear combination of the explanatory variables and their coefficients.

#269 · Generalized Linear Models

Write the formula for the linear predictor η .

Answer

$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$, a linear combination of the predictor variables and their coefficients.

#270 · Generalized Linear Models

What is the link function of a GLM?

Answer

The link function g connects the mean of the response to the linear predictor through $g(\mu) = \eta$, equivalently $\mu = g^{-1}(\eta)$.

#271 · Generalized Linear Models

Why is a link function needed instead of setting $\mu = \eta$ directly?

Answer

The linear predictor η can be any real number, but the mean μ may need to be restricted (for example, always positive or always between 0 and 1). The link function enforces that restriction.

#272 · Generalized Linear Models

What is the canonical link function?

Answer

The canonical link is the specific link function for which the canonical parameter θ equals the linear predictor η exactly, giving mathematically convenient estimation properties.

#273 · Generalized Linear Models

What is the canonical link for the normal distribution?

Answer

The identity link, $g(\mu) = \mu$.

#274 · Generalized Linear Models

What is the canonical link for the Poisson distribution?

Answer

The log link, $g(\mu) = \ln(\mu)$.

#275 • Generalized Linear Models

What is the **canonical link** for the **gamma distribution**?

Answer

The reciprocal link, $g(\mu) = 1/\mu$, though actuaries often use the log link with gamma models instead for interpretability.

#276 • Generalized Linear Models

What is the **canonical link** for the **binomial distribution**?

Answer

The logit link, $g(\mu) = \ln(\mu / (1-\mu))$.

#277 · Generalized Linear Models

Why do actuaries often use the log link with a gamma severity model instead of the canonical reciprocal link?

Answer

The log link gives coefficients a multiplicative interpretation, where each rating factor multiplies the fitted mean by $\exp(\text{beta})$, matching how real rating plans multiply a base rate by relativities.

#278 · Generalized Linear Models

Is using the canonical link mandatory for a GLM?

Answer

No. Any monotonic, differentiable link function can technically be paired with any exponential family distribution, though the canonical link has convenient mathematical properties and is a common default.

#279 · Generalized Linear Models

How is OLS regression a special case of a GLM?

Answer

OLS is the GLM obtained by choosing the normal distribution as the random component and the identity link function, so that the fitted mean equals the linear predictor directly.

#280 · Generalized Linear Models

For the identity link, what is the relationship between μ and η ?

Answer

$\mu = \eta$ directly, since $g(\mu) = \mu$ means no transformation is applied.

#281 • Generalized Linear Models

For the log link, how do you recover the fitted mean μ from the linear predictor η ?

Answer

$\mu = \exp(\eta)$, since $g(\mu) = \ln(\mu) = \eta$ implies exponentiating both sides.

#282 • Generalized Linear Models

For the logit link, how do you recover the fitted mean μ from the linear predictor η ?

Answer

$\mu = \exp(\eta) \div (1 + \exp(\eta))$, the inverse of $g(\mu) = \ln(\mu \div (1-\mu)) = \eta$.

#283 • Generalized Linear Models

A GLM has intercept $\beta_0 = 2,000$ and $\beta_1 = 0.04$ for policy limit, with a normal distribution and identity link. What is the fitted severity for a 100,000 dollar limit?

Answer

$\eta = 2,000 + 0.04 \times 100,000 = 6,000$. Because the link is identity, $\mu = \eta = 6,000$ dollars.

#284 • Generalized Linear Models

A Poisson GLM with a log link has $\beta_0 = -2.50$, $\beta_1 = 0.30$ for youthful driver, $\beta_2 = 0.15$ for urban territory. What is the fitted frequency for a youthful urban driver?

Answer

$\eta = -2.50 + 0.30 + 0.15 = -2.05$. $\mu = \exp(-2.05)$ is about 0.1287 claims per year.

#285 • Generalized Linear Models

In the youthful urban driver example, what is the fitted frequency for a baseline (non-youthful, non-urban) driver?

Answer

$\eta = \beta_0 = -2.50$, so $\mu = \exp(-2.50)$ is about 0.0821 claims per year.

#286 • Generalized Linear Models

In a log-link Poisson model, how do you interpret the ratio of fitted means between two policies that differ only in one indicator variable?

Answer

The ratio equals e raised to the coefficient of that indicator, since fitted means combine multiplicatively under a log link. This gives a relativity interpretation.

#287 • Generalized Linear Models

Why is claim frequency naturally modeled with a Poisson distribution rather than a normal distribution?

Answer

Claim frequency is a count of non-negative integers, and the Poisson distribution is defined exactly on the non-negative integers, unlike the normal distribution which allows any real value.

#288 • Generalized Linear Models

Why is claim severity often modeled with a gamma distribution rather than a normal distribution?

Answer

Severity is strictly positive and right skewed, and the gamma distribution is defined only on positive values and naturally produces right-skewed shapes, unlike the symmetric normal distribution.

#289 · Generalized Linear Models

When is the binomial distribution the appropriate random component for a GLM?

Answer

When the response is a proportion or probability, such as a lapse rate, conversion rate, or a claim indicator across a number of trials.

#290 · Generalized Linear Models

What does it mean that the Poisson distribution has no separate dispersion parameter?

Answer

For the Poisson distribution, $a(\phi)$ is fixed at 1, so the variance is entirely determined by the mean through $\text{Var}(Y) = V(\mu) = \mu$, with no additional scale parameter to estimate.

#291 • Generalized Linear Models

What property of the gamma distribution's variance function explains why percentage variability in severity does not shrink for larger claims?

Answer

Because $V(\mu) = \mu^2$, the standard deviation is proportional to the mean, so the coefficient of variation (standard deviation divided by mean) stays constant across claim sizes.

#292 • Generalized Linear Models

What is the design matrix X in a GLM?

Answer

The design matrix X contains the values of the explanatory variables (rating variables) for every observation, used together with the coefficient vector β to form the linear predictor $\eta = X\beta$.

#293 · Generalized Linear Models

What algorithm is typically used to fit a GLM by maximum likelihood?

Answer

Iteratively reweighted least squares, commonly abbreviated IRLS.

#294 · Generalized Linear Models

In what order are the three GLM components typically chosen when building an insurance pricing model?

Answer

First the random component is chosen to match the nature of the response (Poisson for counts, gamma for severities, binomial for probabilities), then the systematic component is built from candidate rating variables, then the link function is selected.

#295 • Generalized Linear Models

What distribution is sometimes used to model pure premium directly in one step rather than splitting frequency and severity?

Answer

The Tweedie distribution, which can accommodate a mass of exact zeros combined with a continuous positive right-skewed component.

#296 • Generalized Linear Models

What alternative to the Poisson distribution is used when claim count data show overdispersion?

Answer

The negative binomial distribution, which has an additional parameter allowing the variance to exceed the mean.

#297 • Generalized Linear Models

For a normal distribution written in exponential family form with $\theta = \mu$ and $b(\theta) = \theta^2/2$, what is $b'(\theta)$?

Answer

$b'(\theta) = \theta = \mu$, confirming that $E(Y) = \mu$ for the normal distribution.

#298 • Generalized Linear Models

For a normal distribution written in exponential family form with $b(\theta) = \theta^2/2$, what is $b''(\theta)$?

Answer

$b''(\theta) = 1$, a constant, so the variance function $V(\mu) = 1$ and $\text{Var}(Y) = \phi$ (a constant, matching σ^2).

#299 · Generalized Linear Models

For a Poisson distribution written in exponential family form with $\theta = \ln(\mu)$, what is $b(\theta)$?

Answer

$b(\theta) = \exp(\theta) = \mu$.

#300 · Generalized Linear Models

For a Poisson distribution, what are $b'(\theta)$ and $b''(\theta)$ both equal to, and what does that confirm?

Answer

Both equal $\exp(\theta) = \mu$, confirming $E(Y) = \mu$ and $\text{Var}(Y) = \mu$, the defining mean-equals-variance property of the Poisson distribution.

#301 • Generalized Linear Models

What happens to fitted claim frequency predictions under a log link if you compute $\exp(\eta)$ for a very negative η ?

Answer

The fitted mean approaches zero but never becomes negative, since e raised to any real number is always positive, which is exactly why the log link is chosen for count data.

#302 • Generalized Linear Models

Under the logit link, what happens to the fitted probability μ as η approaches positive ∞ ?

Answer

$\mu = \exp(\eta) \div (1 + \exp(\eta))$ approaches 1, staying bounded within the valid range for a probability.

#303 • Generalized Linear Models

Under the logit link, what happens to the fitted probability μ as η approaches negative ∞ ?

Answer

$\mu = \exp(\eta) \div (1 + \exp(\eta))$ approaches 0, staying bounded within the valid range for a probability.

#304 • Generalized Linear Models

Why can a GLM use the same maximum likelihood fitting machinery for Poisson, gamma, binomial, and normal responses?

Answer

Because all four distributions belong to the exponential family, so they share the same general likelihood structure, differing only in their specific $b(\theta)$, $a(\phi)$, and $c(y, \phi)$ functions.

#305 • Generalized Linear Models

What is the practical consequence of using a constant-variance model on data whose true variance grows with the mean?

Answer

The resulting standard errors are wrong, confidence intervals are miscalibrated (too narrow in some ranges, too wide in others), and the parameter estimates are statistically inefficient.

#306 • Generalized Linear Models

In insurance rating, why is a multiplicative model structure (achieved via the log link) often preferred over an additive one?

Answer

Real rating plans are typically built by multiplying a base rate by a series of relativities for each rating variable, and a log link naturally produces this multiplicative structure through fitted coefficients.

#307 · Generalized Linear Models

What does $a(\phi)$ most commonly equal in many GLM applications?

Answer

$a(\phi)$ is very often simply ϕ , or ϕ divided by a known prior weight w , written $a(\phi) = \phi/w$.

#308 · Generalized Linear Models

True or false: a GLM requires the response variable to be normally distributed.

Answer

False. A GLM allows the response to follow any distribution in the exponential family, of which the normal distribution is only one example.

#309 · Generalized Linear Models

What is $c(y, \phi)$ in the exponential family density formula?

Answer

A normalizing term that depends on y and ϕ but does not involve the canonical parameter θ ; it ensures the density integrates or sums to 1.

#310 · Generalized Linear Models

Why can't OLS regression guarantee that fitted claim frequency predictions stay non-negative?

Answer

OLS uses the identity link, so fitted values equal the linear predictor directly, and a linear predictor can take any real value including negative numbers, with nothing to prevent a negative fitted frequency.

#311 • Generalized Linear Models

What variance function does the inverse Gaussian distribution have, relative to Poisson, gamma, and normal?

Answer

The inverse Gaussian has variance function $V(\mu) = \mu^3$, growing even faster with the mean than the gamma distribution's $V(\mu) = \mu^2$.

#312 • Generalized Linear Models

If two GLMs use the same distribution but different link functions, will they generally produce the same fitted coefficients?

Answer

No. The link function changes how the linear predictor relates to the mean, so the estimated coefficients generally differ between link choices even with the same underlying distribution and data.

#313 · Generalized Linear Models

Summarize in one sentence how a GLM generalizes OLS regression.

Answer

A GLM generalizes OLS by allowing the response to follow any exponential family distribution and by connecting the mean to the linear predictor through a chosen link function, with OLS recovered exactly when the distribution is normal and the link is identity.

#314 · Generalized Linear Models

What equation defines the link function g in a GLM?

Answer

$g(\mu) = \eta$, where μ is the mean of the response and η is the linear predictor, $\eta = b_0 + b_1x_1 + b_2x_2 + \dots + b_px_p$.

#315 · Generalized Linear Models

What is the inverse link function used for?

Answer

It converts the linear predictor η back to the mean scale: $\mu = g^{-1}(\eta)$. It guarantees the fitted mean stays within the response variable's valid range.

#316 · Generalized Linear Models

Why does ordinary linear regression (identity link) often fail for insurance response variables?

Answer

It places no restriction on fitted values, so it can predict a negative frequency, a negative severity, or a probability outside 0 and 1, none of which are valid for real insurance outcomes.

#317 · Generalized Linear Models

What is the formula for the identity link?

Answer

$g(\mu) = \mu$, so $\mu = b_0 + b_1x_1 + b_2x_2 + \dots + b_px_p$ directly, with no transformation.

#318 · Generalized Linear Models

Under the identity link, how do predictor effects combine?

Answer

Additively. Increasing x_1 by one unit changes the fitted mean by exactly b_1 , regardless of the values of other predictors.

#319 · Generalized Linear Models

What distribution is typically paired with the identity link?

Answer

The normal (Gaussian) distribution.

#320 · Generalized Linear Models

What is the formula for the log link?

Answer

$g(\mu) = \ln(\mu)$, so $\mu = \exp(\eta) = \exp(b_0 + b_1x_1 + b_2x_2 + \dots)$.

#321 · Generalized Linear Models

Why is the log link standard for Poisson claim frequency models?

Answer

Because $\exp(\eta)$ is always strictly positive for any real η , it guarantees the fitted frequency can never be negative or zero, matching the real-world constraint on claim counts.

#322 · Generalized Linear Models

How does the log link produce a multiplicative rating structure?

Answer

Because $\exp(a+b) = \exp(a) \times \exp(b)$, the fitted mean factors as $\exp(b_0) \times \exp(b_1 x_1) \times \exp(b_2 x_2)$, so each predictor contributes its own independent multiplicative factor to a base rate.

#323 • Generalized Linear Models

What do actuaries call the factors $\exp(\mathbf{b_jx_j})$ produced by a log-link GLM?

Answer

Relativities. Each one adjusts the base rate up or down by a percentage.

#324 • Generalized Linear Models

A Poisson log-link frequency GLM has $b_0 = -2.10$, b_1 (young driver) = 0.30, b_2 (multi-policy) = -0.15. What is the fitted frequency for a young driver with the multi-policy discount?

Answer

$\eta = -2.10 + 0.30 - 0.15 = -1.95$. $\mu = \exp(-1.95) = 0.1423$ claims per exposure unit.

#325 · Generalized Linear Models

In the log-link frequency model with $b_1 = 0.30$ for young driver, what is the young-driver relativity and what does it mean?

Answer

$\exp(0.30) = 1.3499$. A young driver is expected to have about 35.0% more claims than an otherwise identical driver who is not young.

#326 · Generalized Linear Models

What is the formula for the logit link?

Answer

$g(p) = \ln(p \div (1-p))$, the natural log of the odds.

#327 · Generalized Linear Models

What is the inverse logit (logistic) function?

Answer

$$p = \exp(\eta) \div (1 + \exp(\eta)) = 1 \div (1 + \exp(-\eta)).$$

#328 · Generalized Linear Models

Why is the logit link standard for binomial models like lapse or claim occurrence probability?

Answer

The inverse logit function always produces a value strictly between 0 and 1 for any real η , matching the constraint that a probability must fall in that range.

#329 · Generalized Linear Models

As eta approaches negative ∞ under the logit link, what happens to p ?

Answer

p approaches 0.

#330 · Generalized Linear Models

As eta approaches positive ∞ under the logit link, what happens to p ?

Answer

p approaches 1.

#331 • Generalized Linear Models

Under a logit link, what does $\exp(b_j)$ represent?

Answer

The multiplicative change in the odds (not the probability itself) associated with a one-unit increase in x_j .

#332 • Generalized Linear Models

A logit-link lapse model has $b_0 = -1.20$, b_1 (annual pay) = 0.80, b_2 (autopay) = -0.50. What is the lapse probability for a monthly-pay policyholder not enrolled in autopay?

Answer

$\eta = -1.20$. $p = 1 \div (1 + \exp(1.20)) = 1 \div 4.3201 = 0.2315$, about 23.1%.

#333 • Generalized Linear Models

In that same lapse model, what is the lapse probability for an annual-pay, autopay-enrolled policyholder?

Answer

$\eta = -1.20 + 0.80 - 0.50 = -0.90$. $p = 1 \div (1 + \exp(0.90)) = 1 \div 3.4596 = 0.2891$, about 28.9%.

#334 • Generalized Linear Models

Why does a change in the odds scale not translate into the same percentage change on the probability scale under a logit link?

Answer

The logistic function is nonlinear, so a constant multiplicative change in odds produces a different-sized change in probability depending on where the baseline probability sits.

#335 · Generalized Linear Models

What is the formula for the inverse (reciprocal) link?

Answer

$g(\mu) = 1/\mu$, so $\mu = 1/\eta$.

#336 · Generalized Linear Models

What distribution is the inverse link the canonical link for?

Answer

The Gamma distribution, often used for severity.

#337 · Generalized Linear Models

Why do actuaries often prefer the log link over the canonical inverse link for Gamma severity models?

Answer

The log link produces multiplicative relativities that are easier to interpret and easier to combine with a log-link frequency model in a rating plan.

#338 · Generalized Linear Models

What numerical quirk makes the inverse link less stable than the log link?

Answer

Because $\mu = 1/\eta$, the fitted mean is undefined when $\eta = 0$, and the sign of μ flips depending on whether η is positive or negative.

#339 · Generalized Linear Models

An inverse-link Gamma severity model has $b_0 = 0.00040$, b_1 (high-value vehicle) = 0.00010 . What is the fitted severity for a standard-value vehicle?

Answer

$\eta = 0.00040$. $\mu = 1 \div 0.00040 = 2,500$, so \$2,500.

#340 · Generalized Linear Models

In that same inverse-link severity model, what is the fitted severity for a high-value vehicle, and what is notable about it?

Answer

$\eta = 0.00050$. $\mu = 1 \div 0.00050 = 2,000$, so \$2,000. Notably, the larger (more positive) coefficient produced a SMALLER fitted mean, the inverted relationship characteristic of the inverse link.

#341 · Generalized Linear Models

What is an offset in a GLM?

Answer

A term included in the linear predictor whose coefficient is fixed at exactly 1.0 rather than estimated from the data.

#342 · Generalized Linear Models

How is exposure incorporated into a Poisson frequency GLM?

Answer

As an offset equal to $\ln(\text{exposure})$, so the linear predictor becomes $\eta = \ln(\text{exposure}) + b_0 + b_1x_1 + b_2x_2 + \dots$

#343 · Generalized Linear Models

What does including $\ln(\text{exposure})$ as an offset cause the log-link Poisson model to predict?

Answer

The expected claim COUNT, which equals exposure multiplied by a fitted frequency RATE per unit of exposure: $E[\text{claims}] = e \times \exp(b_0 + b_1x_1 + \dots)$.

#344 · Generalized Linear Models

Why must the offset's coefficient be fixed at exactly 1 rather than estimated?

Answer

Fixing it at 1 forces the relationship between exposure and expected claim count to be exactly proportional (doubling exposure exactly doubles expected claims), matching the actuarial definition of frequency as claims divided by exposure.

#345 • Generalized Linear Models

A frequency model with rate 0.1423 claims per car-year is applied to a policy with 0.5 earned car-years exposure using a $\ln(\text{exposure})$ offset. What is the expected claim count?

Answer

$\eta = \ln(0.5) + \text{linear predictor for the rate} = -0.6931 + (-1.95) = -2.6431$. Expected count = $\exp(-2.6431) = 0.0711$, which equals 0.5×0.1423 .

#346 • Generalized Linear Models

What is the risk of including $\ln(\text{exposure})$ as an ordinary predictor with an estimated coefficient instead of a fixed offset?

Answer

The model would not be forced to preserve exact proportionality between exposure and expected claim count, so the fitted rate per unit of exposure would lose its clean multiplicative interpretation.

#347 · Generalized Linear Models

How does a GLM design matrix represent a categorical predictor with k levels?

Answer

With $k-1$ indicator (dummy) variables, each coded 1 if the observation is in that level and 0 otherwise, with one level chosen as the base level that receives no indicator.

#348 · Generalized Linear Models

What happens to the base level's effect in a categorical predictor's design matrix?

Answer

It is absorbed into the intercept b_0 ; every other level's coefficient measures its difference from the base level on the link scale.

#349 · Generalized Linear Models

Does the choice of base level change the fitted values a GLM produces?

Answer

No. It changes how the coefficients are interpreted, but not the fitted values for any combination of predictors.

#350 · Generalized Linear Models

What is an interaction term in a GLM?

Answer

A term equal to the product of two predictors (or their indicators) added to the linear predictor, capturing that the effect of one variable depends on the level of another.

#351 · Generalized Linear Models

How is an interaction term written in the linear predictor?

Answer

$\eta = b_0 + b_1x_1 + b_2x_2 + b_3(x_1 \times x_2)$, where b_3 measures the extra effect from the combination of x_1 and x_2 beyond their separate main effects.

#352 · Generalized Linear Models

What does a GLM assume about two predictors if no interaction term is included between them?

Answer

It assumes their effects are strictly additive on the link scale, meaning the effect of one predictor is the same at every level of the other.

#353 · Generalized Linear Models

Give a classic actuarial example of an interaction term.

Answer

Driver age and vehicle type: a young driver in a sports car may carry substantially more incremental risk than the young-driver effect and sports-car effect would separately suggest.

#354 · Generalized Linear Models

What is multicollinearity?

Answer

A condition where two or more predictors carry overlapping information (are correlated), which can make individual coefficient estimates unstable even though overall model accuracy may still be reasonable.

#355 · Generalized Linear Models

Give an example of two correlated predictors in an auto insurance GLM.

Answer

Vehicle age and vehicle value, which tend to move together.

#356 · Generalized Linear Models

What two criteria should a predictor satisfy before being included in a rating GLM?

Answer

It should be statistically significant and have a plausible, explainable relationship to the outcome, since an unexplainable coefficient is hard to defend to regulators or embed in a live rating plan.

#357 · Generalized Linear Models

Describe the iterative process of GLM model building.

Answer

Fit an initial model, examine diagnostics and residual patterns, test whether adding or removing a predictor or interaction improves the model, and repeat, gradually converging on a model balancing accuracy, interpretability, and stability.

#358 · Generalized Linear Models

Which link function is the canonical link for the Poisson distribution?

Answer

The log link.

#359 · Generalized Linear Models

Which link function is the canonical link for the Binomial distribution?

Answer

The logit link.

#360 · Generalized Linear Models

Which link function is the canonical link for the normal (Gaussian) distribution?

Answer

The identity link.

#361 • Generalized Linear Models

Which link function is the canonical link for the Gamma distribution?

Answer

The inverse (reciprocal) link, $g(\mu) = 1/\mu$.

#362 • Generalized Linear Models

Why is the identity link generally unsuitable for claim frequency modeling?

Answer

It permits negative fitted values, which is impossible for a claim count or frequency, and it does not produce the multiplicative rating structure insurers rely on.

#363 · Generalized Linear Models

On the log scale, how do you convert a fitted coefficient b_j into a relativity?

Answer

Take $\exp(b_j)$. This gives the multiplicative factor by which the fitted mean changes for a one-unit increase in x_j , holding other predictors fixed.

#364 · Generalized Linear Models

What is the base rate in a log-link multiplicative rating structure?

Answer

$\exp(b_0)$, the fitted mean when every predictor is at its base level (all indicators equal 0).

#365 · Generalized Linear Models

True or false: under a log link, relativities for different predictors multiply together onto the base rate.

Answer

True. $\mu = \exp(b_0) \times \exp(b_1x_1) \times \exp(b_2x_2) \times \dots$, so each predictor's relativity multiplies independently onto the base rate.

#366 · Generalized Linear Models

In a lapse GLM with a logit link, does $\exp(\text{coefficient})$ scale the probability or the odds?

Answer

The odds, not the probability directly. The probability itself must be recovered through the full inverse logit transformation.

#367 · Generalized Linear Models

Why is choosing a sensible base level for a categorical predictor useful in practice?

Answer

It makes the fitted coefficients easier to interpret, for example by using the most common or lowest-risk category as the reference point, even though it does not affect the fitted values.

#368 · Generalized Linear Models

What does it mean for exposure to enter a frequency model as a fixed-coefficient term rather than an estimated predictor?

Answer

It means the model is constrained to treat frequency as claims divided by exposure exactly, rather than letting the data determine some other relationship between exposure and claim count.

#369 · Generalized Linear Models

What happens to a Poisson log-link frequency model's fitted rate if exposure doubles, given the offset is correctly specified?

Answer

The fitted rate per unit of exposure stays exactly the same; only the expected claim COUNT doubles, since the offset coefficient is fixed at 1.

#370 · Generalized Linear Models

What business problem is a logit-link GLM typically used for in insurance, besides lapse?

Answer

Modeling any bounded probability outcome, such as probability of claim occurrence, probability of fraud, or probability a quote converts to a bound policy.

#371 · Generalized Linear Models

Why can a coefficient with a plausible sign still be problematic in a GLM with correlated predictors?

Answer

Multicollinearity can make the individual coefficient unstable, so its estimated size (and sometimes even sign) may change substantially with small changes in the data, even if the model's overall fit is fine.

#372 · Generalized Linear Models

What is the general form of the linear predictor η in a GLM with p predictors?

Answer

$\eta = b_0 + b_1x_1 + b_2x_2 + \dots + b_px_p.$

#373 · Generalized Linear Models

Why is the log link considered numerically more stable than the inverse link for practical rating work?

Answer

The log link's inverse transformation, $\exp(\eta)$, is defined and positive for every real η with no discontinuity, unlike the inverse link where $\mu = 1/\eta$ is undefined at $\eta = 0$ and flips sign around it.

#374 · Generalized Linear Models

In model building, why is a single pass at fitting a GLM usually insufficient?

Answer

Because predictor significance, interaction effects, and multicollinearity issues typically only become apparent after examining diagnostics from an initial fit, requiring refinement and re-fitting in cycles.

#375 · Generalized Linear Models

What real-world constraint does the log link's positivity guarantee mirror for severity models?

Answer

A claim severity dollar amount can never be negative or zero, so $\exp(\eta)$ being strictly positive matches that constraint just as it does for frequency.

#376 · Generalized Linear Models

What is the deviance of a fitted GLM?

Answer

Twice the difference between the log-likelihood of the saturated model and the log-likelihood of the fitted model: $D = 2 \times [\log\text{Lik}(\text{saturated}) - \log\text{Lik}(\text{fitted})]$. It measures how far the fitted model falls short of a perfect fit.

#377 · Generalized Linear Models

What is the saturated model?

Answer

The model with one parameter per observation, so its fitted values equal the observed values exactly. It represents the best possible fit achievable given the assumed distribution and is the benchmark deviance is measured against.

#378 · Generalized Linear Models

How does deviance relate to residual sum of squares (RSS) in ordinary regression?

Answer

For a Gaussian GLM with identity link, deviance reduces exactly to RSS. Deviance is described as a generalization of RSS to the broader exponential family (Poisson, gamma, binomial, etc).

#379 · Generalized Linear Models

What is the closed-form deviance formula for a Poisson GLM?

Answer

$D = 2 \times \sum \text{over } i \text{ of } [y_i \times \ln(y_i \div \mu_i) - (y_i - \mu_i)]$, with $y_i \times \ln(y_i \div \mu_i)$ taken as 0 when $y_i = 0$.

#380 · Generalized Linear Models

When is a deviance-based chi-square test valid for comparing two GLMs?

Answer

Only when the two models are NESTED, meaning one is a special case of the other obtained by setting a subset of the larger model's coefficients to zero.

#381 • Generalized Linear Models

For nested Model 0 (p_0 parameters) and Model 1 (p_1 parameters, $p_1 > p_0$), what is the distribution of $\Delta D = D(\text{Model 0}) - D(\text{Model 1})$ under the null hypothesis?

Answer

Approximately chi-square with $p_1 - p_0$ degrees of freedom. This is the likelihood ratio test (LRT) for nested GLMs.

#382 • Generalized Linear Models

In the deviance-based likelihood ratio test, what does it mean if ΔD exceeds the chi-square critical value?

Answer

Reject the null hypothesis: the additional parameters in the larger model improve the fit by more than chance alone would predict, so the larger model is preferred.

#383 · Generalized Linear Models

Can the deviance-based chi-square test compare two non-nested models, such as one using territory and one using population density?

Answer

No. Non-nested models cannot be compared with the deviance chi-square test. Use AIC or BIC instead.

#384 · Generalized Linear Models

What is the formula for the Akaike Information Criterion (AIC)?

Answer

$AIC = -2 \times \logLik + 2 \times k$, where logLik is the maximized log-likelihood and k is the number of estimated parameters.

#385 • Generalized Linear Models

What is the formula for the Bayesian Information Criterion (BIC)?

Answer

$BIC = -2 \times \logLik + k \times \ln(n)$, where k is the number of parameters and n is the number of observations.

#386 • Generalized Linear Models

When comparing candidate models with AIC or BIC, is a lower or higher value preferred?

Answer

Lower is better for both AIC and BIC. The preferred model is the one with the smallest AIC (or smallest BIC).

#387 · Generalized Linear Models

What is the key difference between the AIC penalty and the BIC penalty?

Answer

AIC penalizes each parameter by a constant 2 regardless of sample size. BIC penalizes each parameter by $\ln(n)$, which grows with the number of observations.

#388 · Generalized Linear Models

For what sample sizes does BIC penalize complexity more heavily than AIC?

Answer

Whenever $\ln(n)$ exceeds 2, which happens once n is roughly larger than about 7 or 8 observations. The gap grows as n grows.

#389 · Generalized Linear Models

Does BIC tend to favor simpler or more complex models compared to AIC, especially in large insurance datasets?

Answer

BIC tends to favor simpler, more parsimonious models than AIC, especially as sample size n grows large, since its penalty term $\ln(n)$ increases with n while AIC's penalty stays fixed at 2 per parameter.

#390 · Generalized Linear Models

Can AIC and BIC compare non-nested models?

Answer

Yes. Unlike the deviance chi-square test, AIC and BIC do not require nested models, as long as both models are fit to the identical response variable on the identical set of observations.

#391 · Generalized Linear Models

Is a single AIC value meaningful in isolation?

Answer

No. Only the relative value of AIC (or BIC) across a set of candidate models fit to the same data and response variable is meaningful.

#392 · Generalized Linear Models

Model A has $\log\text{Lik} = -1000$ and 5 parameters. Model B has $\log\text{Lik} = -990$ and 9 parameters, fit on 200 observations. Which does AIC prefer?

Answer

$\text{AIC(A)} = 2000 + 10 = 2010$. $\text{AIC(B)} = 1980 + 18 = 1998$. AIC prefers Model B (lower AIC, $1998 < 2010$).

#393 · Generalized Linear Models

What is forward selection?

Answer

A stepwise procedure that starts with the simplest model (often just an intercept) and, at each step, adds whichever single candidate variable produces the largest improvement in the chosen criterion, stopping when no remaining variable meets the threshold for inclusion.

#394 · Generalized Linear Models

What is backward elimination?

Answer

A stepwise procedure that starts with the full model containing every candidate variable and, at each step, removes whichever variable currently in the model contributes the least to the fit, stopping when every remaining variable meets the threshold for retention.

#395 · Generalized Linear Models

What does stepwise selection (in the narrow sense) do that pure forward or backward selection does not?

Answer

It alternates between forward and backward steps: after each addition it checks whether a previously added variable should now be dropped, and after each removal it checks whether a previously dropped variable should now be re-added.

#396 · Generalized Linear Models

Why are stepwise selection procedures considered unstable?

Answer

Because each step's decision depends on which variables happen to already be in or out of the model, so small changes to the data sample can lead to very different final variable sets.

#397 · Generalized Linear Models

Why can forward selection or backward elimination get stuck in a suboptimal combination of variables?

Answer

Forward selection can never remove a variable once added, and backward elimination can never re-add a variable once removed, so both can miss a combination that a full unconstrained search would have found.

#398 · Generalized Linear Models

Why do stepwise procedures tend to overstate the significance of the final selected variables?

Answer

Because many candidate variables are tested and only the winners are kept, the reported significance levels and standard errors understate the true uncertainty, since the model was effectively chosen to look good on the data used to fit it.

#399 · Generalized Linear Models

What is a key limitation of stepwise procedures regarding interactions and nonlinear terms?

Answer

They do not automatically consider interactions or nonlinear transformations unless those terms are explicitly included in the candidate pool from the start.

#400 · Generalized Linear Models

What is the Pearson residual formula for observation i in a GLM?

Answer

$r_i^{\text{Pearson}} = (y_i - \mu_i) \div \sqrt{V(\mu_i)}$, where μ_i is the fitted mean and $V(\mu_i)$ is the variance function of the assumed distribution.

#401 • Generalized Linear Models

Why is the raw residual ($y_i - \mu_i$) divided by $\sqrt{V(\mu_i)}$ to form the Pearson residual?

Answer

To standardize it so that, under a correctly specified model, Pearson residuals have roughly constant variance across observations, unlike raw residuals whose spread grows with μ_i under Poisson or gamma variance.

#402 • Generalized Linear Models

What property do deviance residuals satisfy in relation to total deviance?

Answer

The sum of the squared deviance residuals across all observations equals the total deviance of the model: $\sum_i (r_i^{\text{deviance}})^2 = D$.

#403 · Generalized Linear Models

Why are deviance residuals generally preferred over Pearson residuals for diagnostic plots?

Answer

Their distribution tends to be closer to symmetric and closer to standard normal under a correctly specified model, making visual patterns easier to interpret.

#404 · Generalized Linear Models

What should a residual plot look like under a correctly specified GLM?

Answer

No systematic pattern: residuals scattered randomly around zero with roughly constant spread when plotted against fitted values or against each predictor.

#405 · Generalized Linear Models

What does a funnel shape in a residual-versus-fitted-values plot suggest?

Answer

That the assumed variance function is wrong, a strong signal to reconsider the distributional assumption or add a dispersion parameter.

#406 · Generalized Linear Models

What does a curved or systematic trend in a residual-versus-predictor plot suggest?

Answer

That the linear predictor is misspecified for that variable, for example that a continuous variable needs a nonlinear transformation or should be binned rather than entered linearly.

#407 · Generalized Linear Models

What is the variance property that defines the Poisson distribution, and why does it matter for GLMs?

Answer

$\text{Var}(Y) = E(Y) = \mu$, meaning variance equals the mean. Insurance claim count data frequently violate this, showing variance exceeding the mean, which is called overdispersion.

#408 · Generalized Linear Models

Name two common causes of overdispersion in Poisson claim count models.

Answer

Unobserved heterogeneity among policyholders (similar observed rating variables but genuinely different underlying claim propensities), and contagion effects (one claim event makes a subsequent claim more likely, violating independence).

#409 · Generalized Linear Models

How is the dispersion estimate ϕ -hat computed to detect overdispersion?

Answer

ϕ -hat = Deviance $\div$ (n - k), where n is the number of observations and k is the number of estimated parameters. Under a correctly specified Poisson model this should be close to 1.

#410 · Generalized Linear Models

What does a dispersion estimate ϕ -hat noticeably greater than 1 indicate?

Answer

That the data show more variability than a plain Poisson model allows for (overdispersion); a common rule of thumb flags roughly 1.5 to 2 or higher as material, though there is no universal cutoff.

#411 · Generalized Linear Models

What is the quasi-Poisson approach to handling overdispersion?

Answer

Keep the Poisson mean structure and coefficient point estimates unchanged, but inflate the standard errors of the coefficients using the estimated dispersion parameter $\hat{\phi}$, correcting inference without changing the fitted means.

#412 · Generalized Linear Models

What is the variance function of the negative binomial distribution, and how does it relate to the Poisson?

Answer

$V(\mu) = \mu + \alpha \times \mu^2$ for dispersion parameter $\alpha > 0$. As α approaches 0, the negative binomial approaches the Poisson. It allows variance to exceed the mean directly.

#413 • Generalized Linear Models

Why is negative binomial regression generally considered a more rigorous fix for overdispersion than quasi-Poisson?

Answer

Because alpha is estimated jointly with the mean-structure coefficients, so the negative binomial model can produce different fitted coefficients (not just different standard errors), modeling the extra variability explicitly.

#414 • Generalized Linear Models

What is the practical risk of ignoring material overdispersion in ratemaking?

Answer

The actuary understates the true uncertainty in the fitted coefficients, which can make unimportant rating variables appear artificially statistically significant.

#415 · Generalized Linear Models

Why do AIC, BIC, and deviance alone fail to detect overfitting?

Answer

They evaluate how well a model fits the same data used to estimate its parameters, so an arbitrarily complex model can always be made to fit the training data better, even by fitting random noise rather than a genuine pattern.

#416 · Generalized Linear Models

What is overfitting?

Answer

When a model fits its training data very well but performs worse than a simpler model on new, unseen data, because it captured noise specific to the training sample rather than the true underlying pattern.

#417 · Generalized Linear Models

What is a single holdout (train÷test) split in cross-validation?

Answer

Randomly dividing the data into a training set (commonly 70 to 80 percent) used to fit the model, and a test set (the remainder) used only to evaluate predictive accuracy after fitting is complete.

#418 · Generalized Linear Models

What is k-fold cross-validation?

Answer

The data is divided into k roughly equal folds (commonly $k = 5$ or 10). The model is fit k times, each time leaving out one fold as the test set, and the k out-of-sample performance measures are averaged.

#419 · Generalized Linear Models

Why is k-fold cross-validation generally preferred over a single holdout split for a limited dataset?

Answer

It makes more efficient use of the data, since every observation is eventually used for testing exactly once, while still ensuring each test observation was excluded from the corresponding training run.

#420 · Generalized Linear Models

How do AIC and BIC relate conceptually to cross-validation?

Answer

AIC and BIC are attempts to approximate what a full cross-validation exercise would show, using only the training data and a mathematical penalty term, rather than actually holding out data and re-fitting.

#421 · Generalized Linear Models

How is a lift chart constructed?

Answer

Policies are scored with the fitted model, sorted from lowest to highest predicted risk, and divided into ten deciles. For each decile, the average predicted loss ratio and the average actual loss ratio are computed and plotted against decile number.

#422 · Generalized Linear Models

What does a monotonically increasing actual loss ratio line across deciles in a lift chart indicate?

Answer

That the model is successfully rank-ordering risk: policies predicted to be worse risks did in fact experience worse actual loss ratios.

#423 · Generalized Linear Models

In a lift chart, what does it mean if the actual and predicted loss ratio lines track closely within each decile?

Answer

That the model's predicted levels, not just its rank ordering, are well calibrated.

#424 · Generalized Linear Models

What does a flat or non-monotonic actual loss ratio line in a lift chart indicate?

Answer

That the model has little or no discriminatory power, meaning it is not successfully separating good risks from bad risks even if its overall average prediction is accurate.

#425 • Generalized Linear Models

What is the Lorenz curve used to build the Gini index?

Answer

Policies are sorted from lowest to highest predicted risk, and the curve plots the cumulative proportion of total actual losses against the cumulative proportion of policies through the sorted order.

#426 • Generalized Linear Models

Where does the Lorenz curve fall if a model has zero discriminatory power?

Answer

Along the 45-degree line of equality, since the bottom x percent of policies by predicted risk would account for exactly x percent of actual losses.

#427 · Generalized Linear Models

How does the Lorenz curve of a model with strong discriminatory power behave?

Answer

It bows well below the line of equality, since the lowest-predicted-risk policies account for a disproportionately small share of actual losses and the highest-predicted-risk policies for a disproportionately large share.

#428 · Generalized Linear Models

How is the Gini index computed from the Lorenz curve?

Answer

As twice the area between the line of equality and the Lorenz curve.

#429 · Generalized Linear Models

What range does the Gini index take, and what does each end mean?

Answer

It ranges from 0 (no discriminatory power beyond random ordering) up toward 1 (very strong discriminatory power; exactly 1 would be perfect separation, not seen with real insurance data).

#430 · Generalized Linear Models

Can the Gini index compare models built on entirely different sets of predictor variables?

Answer

Yes, much like AIC and BIC allow comparison of non-nested models, the Gini index provides a common scale for comparing rank-ordering ability across models with different predictors.

#431 • Generalized Linear Models

What does the Gini index fail to tell you about a model, unlike deviance, AIC, or BIC?

Answer

It says nothing about the calibration or absolute accuracy of the predicted values, only about how well the model orders risks relative to one another.

#432 • Generalized Linear Models

Why should actuaries examine the lift chart and Gini index together with likelihood-based diagnostics rather than relying on one tool alone?

Answer

Because the lift chart and Gini index measure discrimination (rank ordering) while deviance, AIC, and BIC measure fit and calibration; no single tool captures both aspects of model quality.

#433 · Generalized Linear Models

A candidate model has deviance 640 on 200 - 8 = 192 degrees of freedom. Is there evidence of overdispersion?

Answer

$\phi\text{-hat} = 640 \div 192 = 3.33$, well above 1, indicating material overdispersion; the actuary should consider a quasi-Poisson adjustment or a negative binomial model.

#434 · Generalized Linear Models

Two nested Poisson models are compared: Delta D = 15.2 with 2 additional parameters. Is the improvement significant at the 5% level (chi-square critical value with 2 df is 5.99)?

Answer

Yes. $15.2 > 5.99$, so the null hypothesis is rejected and the larger model's additional parameters produce a statistically significant improvement in fit.

#435 · Generalized Linear Models

Why can AIC and BIC disagree on which of two models is better, and when does this become more likely?

Answer

They apply different complexity penalties (2 per parameter for AIC versus $\ln(n)$ per parameter for BIC). Disagreement becomes more likely as sample size n grows, since BIC's penalty grows with $\ln(n)$ while AIC's stays fixed.

#436 · Generalized Linear Models

What is the practical purpose of validating a model on held-out data before it is used for insurance ratemaking?

Answer

To confirm the model's predictive accuracy and discriminatory power generalize to new policies rather than reflecting patterns specific to the training sample, guarding against implementing an overfit model.

#437 · Generalized Linear Models

In a lift chart, why is it important that policies are sorted into deciles using a held-out validation sample rather than the training sample?

Answer

Because in-sample lift and calibration can look artificially strong for an overfit model, so evaluating on held-out data gives a more honest picture of the model's true discriminatory power.

#438 · Generalized Linear Models

What is the essential tradeoff that both AIC and BIC are designed to balance?

Answer

Goodness of fit (captured by $-2 \times \log\text{Lik}$, which decreases as fit improves) against model complexity (captured by a penalty term that grows with the number of parameters k), preventing selection of an overly complex model that merely fits noise.

#439 · Generalized Linear Models

What is the frequency-severity approach to ratemaking?

Answer

Fit two separate GLMs, one for claim frequency and one for claim severity, then multiply the fitted frequency by the fitted severity to get the fitted pure premium for each risk profile.

#440 · Generalized Linear Models

What is the formula for pure premium in terms of frequency and severity?

Answer

Pure Premium = Frequency x Severity, where frequency is expected claim count per exposure and severity is expected dollar amount per claim.

#441 • Generalized Linear Models

Which distribution and link are standard for modeling claim frequency?

Answer

A Poisson (or negative binomial=quasi-Poisson if overdispersed) GLM with a log link.

#442 • Generalized Linear Models

Why is an offset used in a frequency GLM?

Answer

The log of exposure is included as an offset (coefficient fixed at 1.0) so the model predicts a claim rate per unit of exposure rather than a raw claim count.

#443 • Generalized Linear Models

Which distribution and link are standard for modeling claim severity?

Answer

A gamma GLM with a log link, fit only on records that had a claim.

#444 • Generalized Linear Models

Why is the gamma distribution a natural fit for severity data?

Answer

Its variance grows with the square of the mean, matching claim size data where large claims are rare but highly variable and small claims cluster tightly.

#445 • Generalized Linear Models

Does the severity GLM use an exposure offset?

Answer

No. Severity is already a per-claim quantity, so no offset is needed (unlike the frequency GLM).

#446 • Generalized Linear Models

What is the Tweedie distribution used for in ratemaking?

Answer

It models pure premium directly in a single GLM. It is a compound Poisson-gamma distribution: a random Poisson number of claims each with gamma severity, plus a point mass at zero.

#447 · Generalized Linear Models

What does the Tweedie power parameter p represent, and what range does it take for pure premium modeling?

Answer

It controls the variance function shape. $p=1$ recovers the Poisson, $p=2$ recovers the gamma, and values strictly between 1 and 2 blend the two for pure premium modeling.

#448 · Generalized Linear Models

When can the frequency-severity approach and the Tweedie approach diverge meaningfully?

Answer

When a rating variable pushes frequency and severity in opposite directions (for example, younger drivers with higher frequency but lower average severity), a single Tweedie model can mask that offsetting pattern while the two-model approach preserves it.

#449 · Generalized Linear Models

What is one practical drawback of the frequency-severity approach compared to Tweedie?

Answer

The severity model's sample size is much smaller than the frequency model's, since severity data exists only for records that had a claim; it also requires fitting and maintaining two models instead of one.

#450 · Generalized Linear Models

Why does a log-link GLM naturally produce a multiplicative rating structure?

Answer

Exponentiating a linear predictor that is a sum of coefficients turns it into a product of exponentiated terms, so the fitted value equals the base rate times a series of multiplicative relativities.

#451 • Generalized Linear Models

What does $\exp(\beta_0)$ represent in a log-link rating GLM?

Answer

The base rate: the fitted value for a risk that sits in the base level of every rating variable.

#452 • Generalized Linear Models

What does $\exp(\beta)$ represent for a categorical rating variable's coefficient?

Answer

The multiplicative rating relativity for that category relative to the base level. Greater than 1.0 increases the rate; less than 1.0 decreases it.

#453 · Generalized Linear Models

What relativity always applies to the base level of a rating variable, and why?

Answer

A relativity of 1.00, because the base level's dummy variable is dropped from the model and its implicit coefficient is 0, and $\exp(0) = 1$.

#454 · Generalized Linear Models

How do actuaries typically choose the base level for a rating variable?

Answer

They choose the most common or typical category (largest exposure), because it produces a more stable intercept estimate and makes relativities easier for underwriters and regulators to interpret.

#455 · Generalized Linear Models

A frequency GLM has intercept $\beta_0 = -2.30$. What is the base rate?

Answer

$\exp(-2.30) = 0.1003$ claims per exposure.

#456 · Generalized Linear Models

A territory coefficient is $\beta_T = 0.2400$. What is the relativity and what does it mean?

Answer

$\exp(0.2400) = 1.271$. That territory is rated 27.1% higher than the base territory.

#457 • Generalized Linear Models

A vehicle age coefficient is $\beta_V = -0.1050$. What is the relativity and what does it mean?

Answer

$\exp(-0.1050) = 0.900$. That vehicle age band is rated 10.0% lower than the base age band.

#458 • Generalized Linear Models

Given a base rate of 0.1003, a territory relativity of 1.271, and a vehicle age relativity of 0.900, what is the fitted frequency for that risk?

Answer

$0.1003 \times 1.271 \times 0.900 = 0.1147$ claims per exposure.

#459 · Generalized Linear Models

If fitted frequency is 0.1147 and fitted severity is \$4,800, what is the fitted pure premium?

Answer

$$0.1147 \times \$4,800 = \$550.56.$$

#460 · Generalized Linear Models

How many coefficients does a categorical rating variable with k levels contribute to a GLM?

Answer

k-1 coefficients, since one level is held out as the base level.

#461 • Generalized Linear Models

Why is territory difficult to model as an ordinary unpenalized categorical variable?

Answer

It often has hundreds of distinct levels (ZIP codes or territory codes), leading to thin cells, unstable coefficient estimates, and severe overfitting.

#462 • Generalized Linear Models

What is territorial clustering?

Answer

Grouping raw geographic units into a manageable number of territories based on preliminary loss experience analysis (such as univariate relativities or a clustering algorithm) before feeding the groups into the GLM.

#463 · Generalized Linear Models

What is spatial smoothing in territory ratemaking?

Answer

A technique such as a spatial random effect that borrows statistical strength from neighboring geographic areas, pulling a thinly populated territory's rate toward its neighbors rather than relying only on its own limited data.

#464 · Generalized Linear Models

What should an actuary do with a sparse level of a classification variable that has too little exposure?

Answer

Group it with a neighboring, similar level before final model fitting, rather than leave it to produce an unstable, statistically insignificant relativity.

#465 • Generalized Linear Models

What is the tradeoff between treating a numeric rating variable as smooth÷continuous versus banding it into a categorical variable?

Answer

Banding is simpler to implement and communicate but sacrifices statistical efficiency and can create artificial rate discontinuities, while a smooth curve is more efficient but harder to administer.

#466 • Generalized Linear Models

What is a common compromise for choosing band boundaries for a continuous rating variable?

Answer

Fit the variable with a smooth curve first, inspect its shape, then choose band boundaries that approximate the smooth curve while keeping the final rating table administratively simple.

#467 · Generalized Linear Models

On what scale is a GLM coefficient reported in a typical coefficient table?

Answer

On the link scale, which for a log-link model means the log scale, not the original (dollar or count) scale.

#468 · Generalized Linear Models

What does the standard error of a GLM coefficient measure?

Answer

How precisely the coefficient has been estimated. It shrinks with more exposure in the relevant category and grows when exposure is thin or the response is highly variable.

#469 · Generalized Linear Models

What is the common rule of thumb for statistical significance of a GLM coefficient at roughly the 95% confidence level?

Answer

An absolute z-statistic (coefficient divided by its standard error) exceeding about 1.96, equivalently a p-value below 0.05.

#470 · Generalized Linear Models

What is the formula for an approximate 95% confidence interval for a GLM coefficient beta?

Answer

$\text{beta} \pm 1.96 \times \text{SE}(\text{beta})$.

#471 • Generalized Linear Models

How do you convert a confidence interval for a coefficient into a confidence interval for the relativity?

Answer

Exponentiate both endpoints: $\exp(\text{beta} - 1.96 \times \text{SE}(\text{beta}))$ to $\exp(\text{beta} + 1.96 \times \text{SE}(\text{beta}))$.

#472 • Generalized Linear Models

A severity coefficient is $\text{beta} = 0.1823$ with $\text{SE} = 0.0410$. What is the z-statistic?

Answer

$0.1823 \div 0.0410 = 4.45$, well above the 1.96 threshold, so highly statistically significant.

#473 · Generalized Linear Models

For $\beta = 0.1823$ and $SE = 0.0410$, what is the approximate 95% confidence interval for β ?

Answer

$0.1823 \pm 1.96 \times 0.0410 = 0.1823 \pm 0.0804$, or (0.1019, 0.2627).

#474 · Generalized Linear Models

For the confidence interval (0.1019, 0.2627) on the log scale, what is the confidence interval for the relativity?

Answer

$\exp(0.1019) = 1.107$ to $\exp(0.2627) = 1.301$, with a point estimate relativity of $\exp(0.1823) = 1.200$.

#475 · Generalized Linear Models

Can a coefficient be statistically significant but practically unimportant? How?

Answer

Yes. In a very large dataset, a trivially small effect (for example, a relativity of 1.01) can be statistically distinguishable from 1.00 even though it has essentially no real-world pricing consequence.

#476 · Generalized Linear Models

Why might an actuary retain a rating variable that is not statistically significant in the current dataset?

Answer

If it is a well-established rating factor supported by industry experience or prior studies, and the actuary judges the current sample is simply too thin to detect a known real effect.

#477 · Generalized Linear Models

What is the difference between statistical significance and practical (business) significance?

Answer

Statistical significance means an effect is distinguishable from zero given the data's precision; practical significance means the effect is large enough in magnitude to matter for actual pricing decisions.

#478 · Generalized Linear Models

What is a holdout (out-of-sample) test set used for in ratemaking model validation?

Answer

Data held out entirely from model fitting, used to measure the model's predictive accuracy on data it has never seen, to detect overfitting.

#479 · Generalized Linear Models

What does it mean if a model performs well on training data but noticeably worse on test data?

Answer

It is a strong signal of overfitting: the model captured noise specific to the training sample rather than a genuine underlying pattern.

#480 · Generalized Linear Models

What are two common metrics used to measure predictive accuracy on holdout data?

Answer

Deviance and the Gini index, computed on the held-out test set rather than the training data.

#481 • Generalized Linear Models

What is an actual-versus-expected (A÷E) comparison in ratemaking validation?

Answer

For each level of each rating variable, comparing actual observed losses÷claim counts÷pure premium in holdout data against what the model predicted for that same group.

#482 • Generalized Linear Models

What should A÷E ratios look like for a well-calibrated rating model?

Answer

Close to 1.00 across essentially every level of every rating variable.

#483 • Generalized Linear Models

What does a persistent, large A-E deviation in one territory suggest?

Answer

The model is missing an important interaction or nonlinearity, or the rating variable needs further refinement (additional banding, an interaction term, or spatial smoothing).

#484 • Generalized Linear Models

Why is overfitting especially dangerous in a ratemaking context, beyond just poor statistical fit?

Answer

An overfit rating plan misprices real policyholders, systematically overcharging some segments and undercharging others, eroding both profitability and competitiveness.

#485 • Generalized Linear Models

Why is a rating variable like territory especially prone to overfitting?

Answer

It has many levels with thin exposure in some, so a handful of large claims randomly concentrated in one small territory can produce an extreme, unstable relativity unrelated to true underlying risk.

#486 • Generalized Linear Models

What is cross-validation used for in ratemaking model validation?

Answer

Repeatedly splitting data into different training÷test partitions and averaging model performance across the splits, to detect instability in indicated relativities before a rating plan goes live.

#487 • Generalized Linear Models

What is credibility weighting used for once an unstable relativity is identified?

Answer

Correcting the instability by weighting the raw indicated relativity toward a broader base rate, giving less weight to thin, noisy segments.

#488 • Generalized Linear Models

Why must a GLM-based rating plan usually be filed with a regulator before use?

Answer

In most jurisdictions the plan must be approved by an insurance regulator, and the actuary must provide clear actuarial justification for every rating variable and relativity.

#489 · Generalized Linear Models

What does actuarial justification for a rating variable generally require?

Answer

Demonstrating statistically significant coefficients, stable relativities across cross-validation folds or time periods, sensible A÷E performance on holdout data, and a plausible connection to loss potential.

#490 · Generalized Linear Models

Are regulators generally comfortable with a rating variable that has a purely statistical relationship to risk with no plausible causal or correlational story?

Answer

No. Regulators are typically skeptical of such a variable, and the actuary should be prepared to explain in plain terms why it is expected to relate to risk.

#491 • Generalized Linear Models

Name examples of variables commonly prohibited or restricted in insurance rating.

Answer

Race, religion, national origin, and in many jurisdictions and lines, credit-based insurance scores, gender, or genetic information (the specific list varies by jurisdiction and line of business).

#492 • Generalized Linear Models

What is a proxy variable risk in GLM ratemaking?

Answer

A GLM can pick up statistically significant relationships with variables highly correlated with a prohibited characteristic even when that characteristic is never directly included in the model; actuaries must check for and exclude this indirect effect.

#493 • Generalized Linear Models

Is statistical significance alone sufficient to satisfy a regulator reviewing a rate filing?

Answer

No. The actuary must also show the resulting relativity is reasonable in magnitude, consistent with prior experience or industry benchmarks, and does not cause excessive rate disruption.

#494 • Generalized Linear Models

What is rate disruption, and why do many jurisdictions cap it?

Answer

A large swing in an individual policyholder's premium from a rating plan change. Many jurisdictions cap how much a single revision can move an individual's premium in one renewal cycle, for fairness and market-stability reasons.

#495 • Generalized Linear Models

What two things must an actuary balance when translating raw GLM-indicated relativities into a final implementable rating plan?

Answer

The pure statistical evidence from the model and regulatory constraints such as rate disruption caps and prohibited-variable rules.

#496 • Generalized Linear Models

In a log-link GLM, why is the combined pure premium from frequency times severity automatically multiplicative in the rating factors?

Answer

Because both fitted values are exponentials of linear predictors, so multiplying them multiplies the exponentiated coefficient terms together, preserving the multiplicative base-rate-times-relativities structure.

#497 • Generalized Linear Models

What administrative advantage does a single Tweedie pure premium model have over the two-model frequency-severity approach?

Answer

It requires fitting and maintaining only one model instead of two, and avoids the sample-size mismatch problem between frequency data (all exposures) and severity data (only claims).

#498 • Generalized Linear Models

Why do actuaries examine frequency and severity separately even when a Tweedie model could combine them?

Answer

Separate models give distinct insight for underwriting or claims purposes, such as which rating variables drive claim frequency versus which drive claim cost.

#499 • Generalized Linear Models

What happens to a rating variable's coefficient standard error as its exposure (sample size) in that category increases?

Answer

The standard error shrinks, producing a more precisely estimated coefficient and relativity.

#500 • Generalized Linear Models

Why is it important to check A÷E performance broken out by rating variable rather than relying only on an aggregate fit statistic like overall deviance?

Answer

An aggregate statistic can look good even while the model performs poorly for specific segments; breaking out A÷E by rating variable level reveals localized miscalibration that aggregate metrics can hide.